



Tribhuvan University
Faculty of Humanities and Social Science

A PROJECT REPORT ON
SOCIAL MEDIA AND SOCIAL NETWORKING SYSTEM (MITRAPARK)

Submitted to
Department of Computer Application
Ratna Rajyalaxmi Campus
Pradarshani Marg, Kathmandu

In partial fulfillment of the requirements for the Bachelor in Computer Application

Submitted by
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Supervisor's Recommendation

I hereby recommend that this project prepared under my supervision by Alen Pariyar entitled “**SOCIAL MEDIA AND SOCIAL NETWORKING SYSTEM (MITRAPARK)**” in partial fulfilment of the requirements for the degree of Bachelor of Computer Application is recommended for the final evaluation.

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LETTER OF APPROVAL

This is to certify that this project prepared by Alen Pariyar entitled “**SOCIAL MEDIA AND SOCIAL NETWORKING SYSTEM (MITRAPARK)**” in partial fulfilment of the requirements for the degree of Bachelor of Computer Application has been evaluated. In our opinion, it is satisfactory in the scope and quality of a project for the required degree.

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Abstract

In the competitive world of social media, having a strong online presence is important. This project focuses on creating MitraPark, a social networking website where people can connect, share, and interact with each other. MitraPark uses simple technologies like HTML (Hypertext Markup Language), CSS (Cascading Style Sheet), and JavaScript for the website's look and feel, and PHP (Hypertext Pre-processor) and MySQL to handle data and make things work behind the scenes. The system consists of web pages accessible through a web browser.

MitraPark offers various features like user profiles, posts, comments, likes, friend management, messaging to make the experience enjoyable for users. Additionally, the project includes documentation covering project planning, analysis, design, implementation, and testing. MitraPark aims to provide users with a welcoming space to express themselves, connect with others, and build relationships online.

Keywords: *social media, social networking, website, HTML, CSS, JavaScript, PHP, MySQL, user profiles, comments, likes, comments, friend management, messaging, documentation.*

Acknowledgment

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May 24, 2024

Alen Pariyar (T.U. Registration No. 6-2-40-4-2021)

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List of Abbreviations

AJAX	-	Asynchronous JavaScript and XML
CASE	-	Computer-Aided Software Engineering
CSS	-	Cascading Style Sheets
DFD	-	Data Flow Diagram
ER	-	Entity Relationship
HTML	-	Hypertext Markup Language
JS	-	JavaScript
PHP	-	Hypertext Pre-processor
SQL	-	Structured query language
UI	-	User Interface

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Chapter 1 : Introduction

1.1. Introduction

In today's digital age, social media and social networking systems have become integral parts of our daily lives, fundamentally altering the way we communicate, connect, and interact with others. Social media platforms enable users to share information, engage in conversations, and build relationships with individuals across the globe. Likewise, social networking systems provide features for users to create personal profiles, connect with friends around shared interests and activities. [1]

MitraPark is a dynamic and innovative social networking platform crafted to provide a simple platform for online social interaction. Serving as a virtual hub for connectivity and engagement, MitraPark offers users a multifaceted platform to connect with friends, share experiences, and discover new content. [2]

Unlike existing social media platforms, MitraPark distinguishes itself by offering an ad-free and sponsorship-free environment, providing users with a straightforward and authentic platform for social media and networking. [3] MitraPark prioritizes genuine connections and seamless connectivity, ensuring that users can engage with content and interact with others without distractions or interruptions. By maintaining a simple yet effective approach, MitraPark aims to create a space where users can truly connect and engage in meaningful interactions without the influence of external advertisements or sponsored content. [4]

1.2. Problem Statement

In existing social media platforms there is no such functionalities to make control over the maximum accessible time. Due to that user can use the system for unset time boundary and as result they are being addicted to social media platforms and loosing their productivity in personal life also often prioritize advertising revenue and sponsored content over user experience, leading to a very crowded and annoying online environment. Users often face challenges in maintaining genuine connections and authentic interactions due to influence of advertisements and promotional content. To address these issues, there is a pressing need for a new social networking platform that prioritizes user privacy, authenticity, and simplicity. MitraPark aims to fill this gap by providing users with an ad-free and sponsorship-free platform that fosters genuine connections, facilitates meaningful interactions.

1.3. Objectives

The objective for developing MitraPark is to create a dynamic and user-centric social media platform that addresses the shortcomings of existing systems and fulfils the following key objectives:

- To develop a web-based social media system that provides a platform for users to connect with other users,
- To provide users the features to post, message, and interact with each other

1.4. Scope and Limitation

Scope

Being a Social Media and Networking site it has a great future scope. As a web-based system it can be used with all web browser enabled Operating Systems. With the increasing reliance on online connectivity and communication, the demand for user-friendly and simple platforms like MitraPark is expected to rise.

Limitations

It is focused on simplicity amidst a landscape dominated by highly competitive social media platforms, MitraPark faces several challenges:

- **Market Competition:** MitraPark enters a crowded market with established players, making it challenging to carve out a niche and attract users away from existing platforms.
- **User Adoption:** Encouraging users to switch from familiar platforms to MitraPark may be difficult, requiring compelling features and incentives to drive adoption.
- **Resource Constraints:** Limited resources, including funding and manpower, may hinder MitraPark's ability to compete effectively with larger, well-established competitors.
- **Feature Parity:** MitraPark's emphasis on simplicity may result in a lack of advanced features compared to competitors, potentially limiting its appeal to certain user segments.
- **Brand Recognition:** MitraPark may struggle to gain visibility and recognition among users accustomed to established social media brands, requiring significant marketing efforts to build awareness and trust.

Despite these challenges, MitraPark can leverage its unique points, such as simplicity and user-focused design, to differentiate itself and attract the audience. Additionally, innovative marketing strategies, and continuous refinement of its platform based on user feedback can

help MitraPark overcome these limitations and establish a foothold in the competitive social media landscape.

1.5. Report Organization

The first part of the report contains the summarized introduction of the whole report. It includes the overview, scope and limitation, problem statement and objectives of this project.

The second chapter includes a background study i.e. description of fundamental theories, general concepts, and terminology related to the project. It also includes the literature review i.e., a review of similar projects, research, and theories done by other researchers.

The third chapter includes the system analysis and design phase in which the report of functional and non-functional requirements of the project is stated using use cases and system diagrams. It also includes the feasibility study about the system which explains whether the system development process is affordable and within the knowledge range of the developers. It shows the technical, operational, and economic feasibility of the project development phase. The explanation of the design of the system is also done in this chapter. It includes data modelling and process modelling which is explained by using ER diagram and Data Flow Diagram. The architectural design, database design, and user interface design are also listed in this chapter.

The fourth chapter includes the implementation and testing phase of the proposed system. In the implementation phase, tools like CASE tools, programming languages and database platforms are implemented.

The fifth chapter includes a conclusion and future recommendations. This contains the final paragraphs of the report and in this phase the overall outcome and the developer's point of view.

Chapter 2: Background Study and Literature Review

2.1. Background Study

The background study delves into fundamental theories, general concepts, and terminologies pertinent to the project, providing a foundational understanding of key elements shaping the development of MitraPark.

In the context of social media and networking systems, several fundamental theories and concepts underpin the design and functionality of platforms like MitraPark. One such concepts such as user engagement, user experience (UX), and user interface (UI) design play crucial roles in shaping the effectiveness and appeal of social networking platforms. User engagement refers to the degree of interaction and participation exhibited by users on the platform, while UX and UI design focus on creating intuitive and visually appealing interfaces that enhance user satisfaction and usability.

Furthermore, terminology related to social networking systems, such as user profiles, posts, comments, likes, and privacy settings, forms the building blocks of MitraPark's functionality. User profiles serve as digital representations of individuals on the platform, while posts enable users to share content with their network. Comments and likes facilitate engagement and interaction.

2.2. Literature Review

We researched and reviewed some of the related websites, and articles in this project. By reviewing existing literature and studies, MitraPark gains valuable insights into user behaviour, platform design principles, and emerging trends in social media usage. This knowledge serves as a foundation for informed decision-making and strategic planning, ensuring that MitraPark effectively meets the needs and expectations of its users in a rapidly evolving digital landscape.

Deitel's "Internet & WorldWideWeb How To Program" serves as a comprehensive guide to understanding the fundamentals of internet programming and web development. This textbook provides a thorough introduction to HTML, CSS, JavaScript, and various other web technologies, making it an essential resource for both beginners and experienced developers. Its structured approach covers both theoretical and practical aspects, enabling readers to build dynamic and interactive web applications. The book also delves into the history and evolution of the internet, offering insights into how web standards and protocols have developed over time. [1]

Facebook, a social networking platform developed by Meta, has revolutionized the way people connect and communicate online. Launched in 2004, Facebook has grown to become one of the most popular social media platforms worldwide, with billions of active users. It allows individuals and organizations to create profiles, share content, interact through likes, comments, and messages, and join groups based on common interests. Facebook's role in social networking extends beyond personal interactions to include business marketing, community building, and political activism, highlighting its significant impact on modern digital communication. [5]

Instagram, another social media platform under Meta, focuses on photo and video sharing. Since its launch in 2010, Instagram has gained immense popularity, especially among younger demographics. Users can post photos and videos, apply various filters, and share their content with followers. Instagram Stories, IGTV, and Reels are notable features that have enhanced user engagement by offering different formats for content creation and sharing. The platform has also become a vital tool for influencers and businesses, leveraging its visual-centric approach for marketing and brand promotion. [2]

S. M. Musa's publication on social networking explores the evolution and impact of social media platforms on society. The paper examines how social networking sites have changed the way individuals communicate, form relationships, and engage in community activities. It also addresses the potential risks and challenges associated with social networking, such as privacy concerns, cyberbullying, and the spread of misinformation. Musa's analysis provides a critical overview of the social, psychological, and cultural implications of social media usage, contributing to a deeper understanding of its role in contemporary life. [3]

Twitter, now rebranded as X, is a microblogging platform that allows users to post and interact with short messages called tweets. Launched in 2006, Twitter has become a powerful tool for real-time communication, news dissemination, and public discourse. Its unique format, limited to 280 characters per tweet, encourages concise and impactful communication. The platform's influence extends across various domains, including media, entertainment, and academia. [6]

Ownmates is a social media platform designed to connect people with similar interests and hobbies. It aims to create a more personalized and meaningful online community experience by matching users based on shared activities and preferences. Ownmates offers various features such as group chats, event planning, and collaborative projects, fostering

a sense of belonging and collaboration among its users. By focusing on common interests, Ownmates differentiates itself from other social networks, emphasizing quality interactions over quantity and providing a niche space for like-minded individuals. [7]

E. O. Femi Odusote's work details the development of a social media-based web application aimed at prospective university students. The application serves as a platform for students to interact, share information, and seek guidance during the university selection process. It includes features like discussion forums, information resources, and peer-to-peer support, enhancing the pre-university experience. The case study of Covenant University highlights the practical aspects of designing and implementing such a platform, emphasizing the importance of user-friendly interfaces and relevant content to engage and support students effectively. [8]

H. H. H. Innocencia Owuor's overview explores the integration of social media apps in geospatial research. The paper discusses how platforms like Facebook, Twitter, and Instagram can be leveraged for collecting and analyzing geospatial data. Social media's real-time data generation and widespread usage make it a valuable resource for mapping trends, monitoring environmental changes, and understanding human mobility patterns. Owuor highlights the potential of social media to enhance geospatial research methodologies, providing new avenues for data collection and analysis in various scientific and practical applications. [4]

A. S. Alfonso Pellegrino's bibliometric analysis investigates the growing research trends in social media addiction and problematic social media use. The study identifies key themes, methodologies, and findings from recent literature, highlighting the psychological, social, and behavioral impacts of excessive social media use. It discusses factors contributing to addiction, such as the design of social media platforms and individual susceptibility. The analysis also addresses the consequences of social media addiction, including mental health issues, reduced productivity, and interpersonal conflicts, emphasizing the need for effective interventions and public awareness. [9]

T.-H. L. Muhammad Moiz Khan's literature review examines the causes and consequences of social media addiction. The review synthesizes findings from various studies, identifying factors such as personality traits, social influences, and platform-specific features that contribute to addictive behaviors. It also explores the adverse effects of social media addiction, including mental health disorders, academic decline, and social isolation. Khan's

work underscores the complexity of social media addiction, calling for comprehensive strategies to mitigate its negative impact on individuals and society. [10]

By assimilating lessons from these prominent platforms, MitraPark strives to combine the best elements from existing platforms to create a unique and user-centric social media experience.

Chapter 3: System Analysis and Design

3.1. System Analysis

3.1.1. Requirement Analysis

i. Functional Requirements

A functional requirement is a description of the functionalities of Social Media and Social Networking Sites. The system's features are detailed through the creation of use case diagrams, which help to effectively represent the functional requirements of the Social Media and Social Networking Site(MitraPark).

The system provides the following functionalities:

User Module:

- Users can Register into the system
- Users can Login / Logout into the system
- Users can manage posts
- Users can send messages
- Users can like in the post
- User can comment on post
- User can update profile
- Users are able to send and receive friend requests

Admin Module

- Admin can Login / Logout
- Admin can manage users
- Admin can manage posts

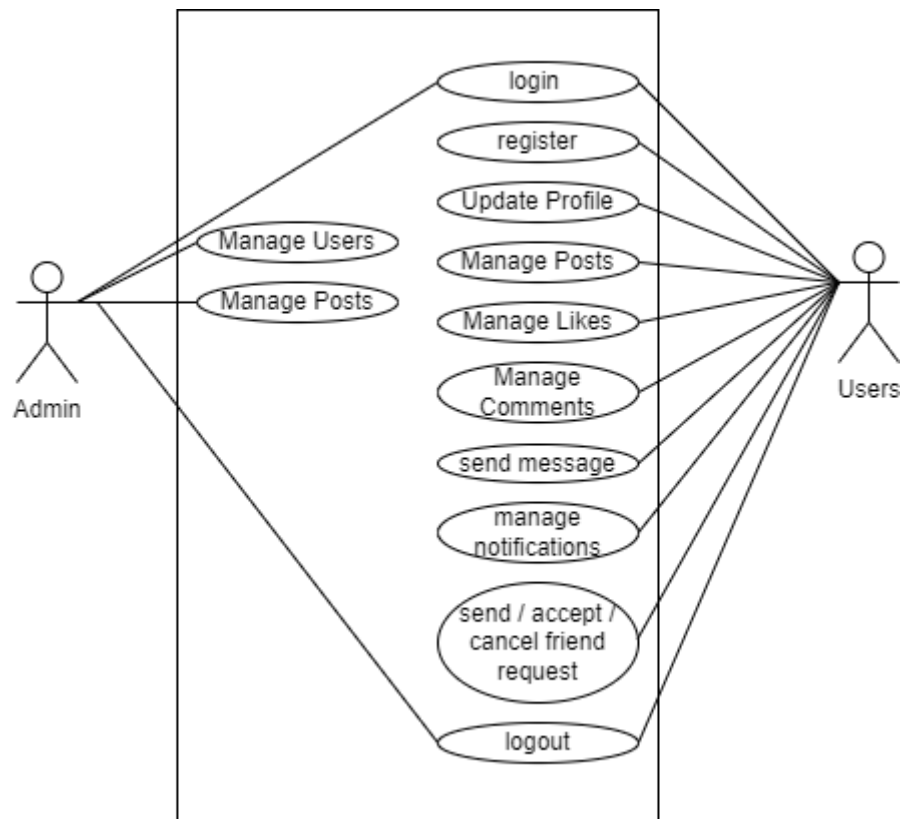


Figure 3.1: Use-case diagram for Social Media and Networking System 'MitraPark'

These functional requirements can be represented as following in the USE CASE Diagram:

ii. Non-functional Requirements

Non-functional requirements, also known as quality attributes or system qualities, define the characteristics that describe how a system should behave, rather than what the system should do. These requirements address aspects such as performance, security, usability, reliability, and scalability. For MitraPark, some non-functional requirements may include:

- **Performance** - MitraPark should have fast response times for loading pages and displaying content to ensure a smooth user experience.
- **Security** - The platform should have mechanisms in place to prevent unauthorized access, such as strong password requirements and session management controls.
- **Usability** - MitraPark should have an intuitive and user-friendly interface that is easy to navigate and understand, catering to users of all levels of technical proficiency.
- **Reliability** - MitraPark should be highly available and reliable, with minimal downtime and disruptions to service.
- **Compatibility** - MitraPark should be compatible with a wide range of web browsers and devices to ensure a consistent user experience across different platforms.

These non-functional requirements are essential for ensuring that MitraPark delivers a high-quality, reliable, and user-friendly experience to its users while also addressing critical aspects such as performance, security, and combability.

3.1.2. Feasibility Study

The analysis of feasibility has concluded that the project is feasible with respect to time and cost. The technology used to develop is almost Open Source, therefore less cost for implementation and maintenance will be involved. Feasibility is the study of impact, which happens in the organization by the development of a system. The impact can be either positive or negative. When the positives nominate the negatives, then the system is considered feasible. Here the feasibility study has been done in three ways:

i. Technical Feasibility

The system is currently implemented using readily available technologies, ensuring both present and future technical feasibility. Being web-based, it operates seamlessly on any system equipped with a web browser, eliminating the need for specific hardware or software requirements. The system utilizes HTML, CSS, and JavaScript for the frontend, while PHP and MySQL handle the backend, providing users with a robust and efficient platform. This combination of technologies ensures compatibility, ease of access, and a smooth user experience across diverse devices and operating systems.

ii. Operational Feasibility

Operational feasibility is all about how well the system solves the problem and takes advantage of identified opportunities during the scope definition of the system. The MitraPark project is operationally feasible, ensuring its practicality and effectiveness in the social media landscape. Through a thorough examination of the current mode of operation, the project ensures that it aligns seamlessly with existing processes, providing enhanced throughput and response times. The proposed system of MitraPark is designed to genuinely benefit individuals, offering improvements to current work practices and procedures within the social networking realm. Operational feasibility is further confirmed through an analysis of resource utilization, including people, time, and the flow of forms, ensuring that MitraPark makes optimal use of available resources for a streamlined user experience.

iii. Economic Feasibility

The system is economically feasible as it does not require any features to be purchased or premium subscriptions to be taken. All functionalities are available to users at no

additional cost, making the system accessible without financial barriers. This ensures that users can fully utilize the system's capabilities without incurring any extra expenses, promoting wide adoption and cost-effectiveness.

iv. Schedule Feasibility

This includes the project schedule and all time allocated for their completion. The Gantt chart is as follows:

MitraPark																
Months	Poush 2080				Magh 2080				Falgun 2080				Chaitra			
Week	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
Planning																
Analysis																
Design																
Coding																
Testing																
Documentation																

Figure 3. 2: Gantt Chart

3.1.3. Data Modelling

Data modeling is the process of designing the structure of a database system to represent the information and relationships between different entities in the domain of interest. It involves identifying the entities, attributes, and relationships within the system and organizing them in a way that accurately reflects the real-world scenario being modeled. ER (Entity-Relationship) Model is mostly used for data modeling.

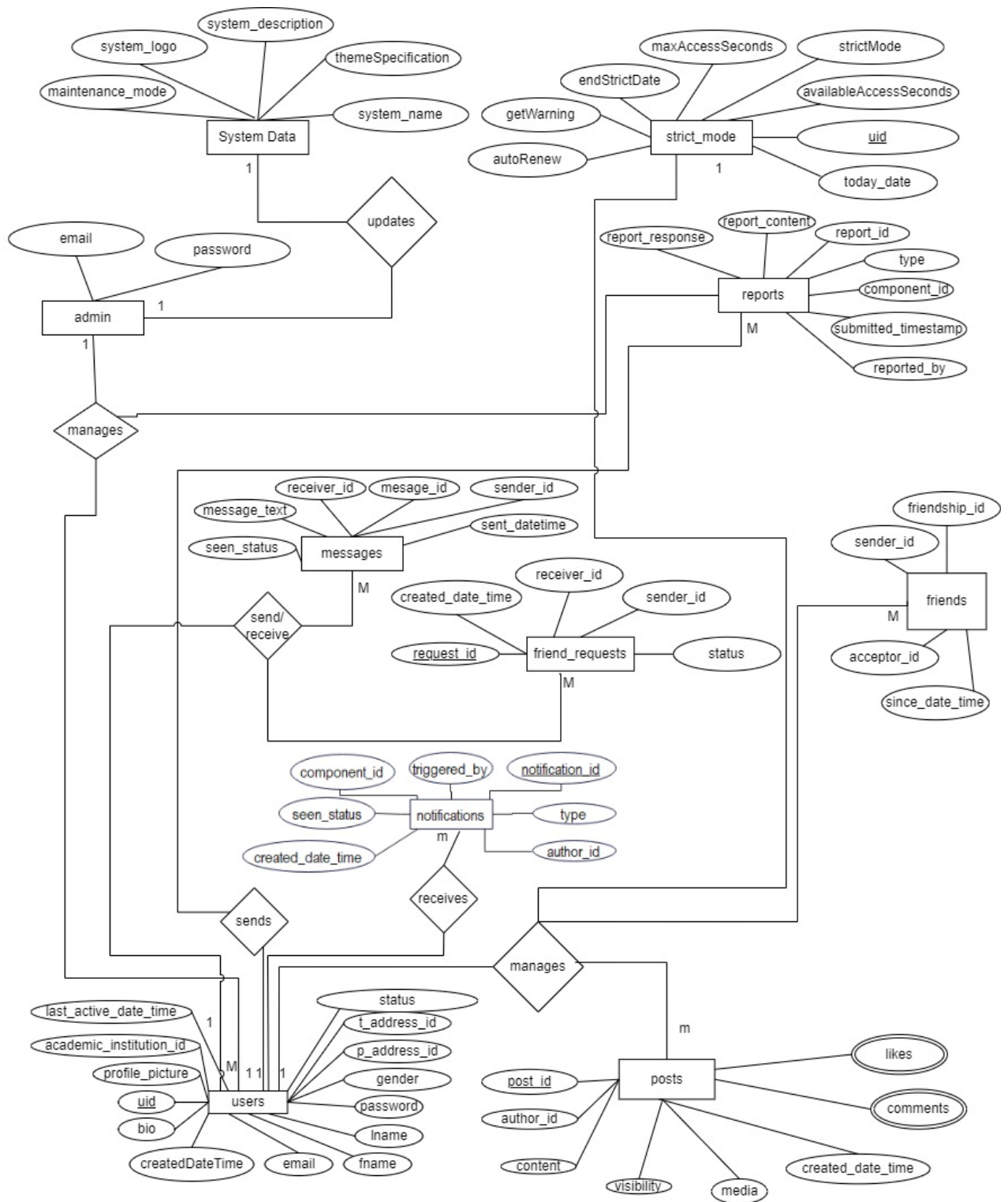


Figure 3.3: ER-Diagram

3.1.4. Process Modeling (DFD)

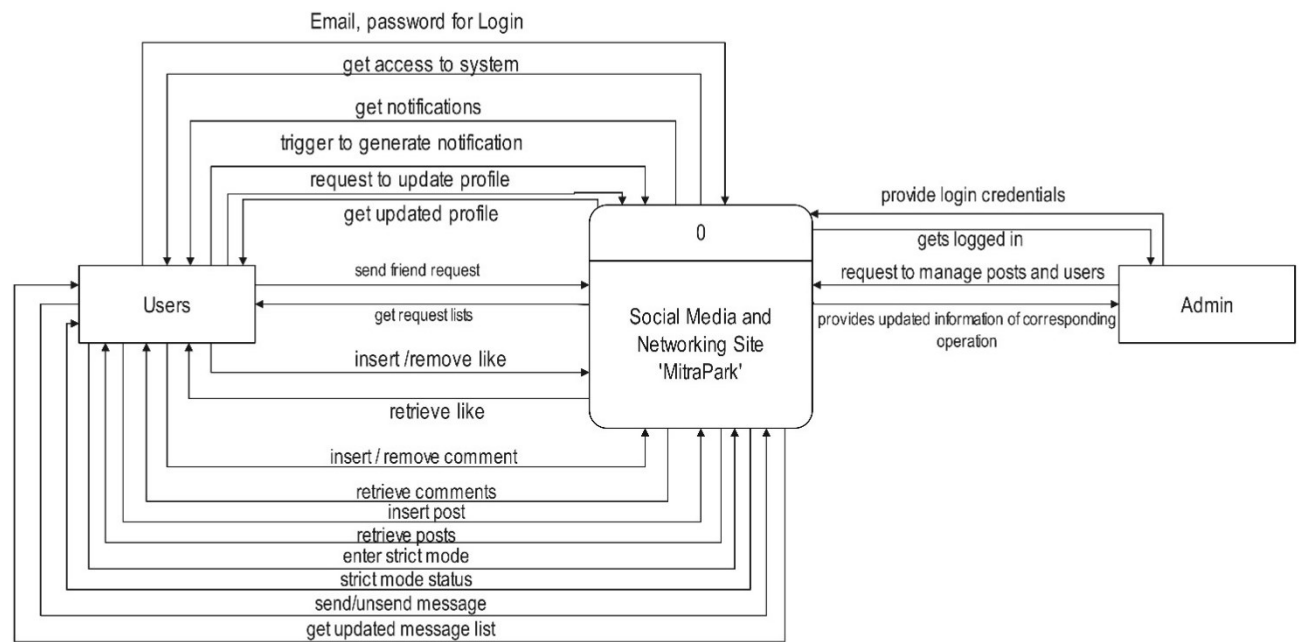


Figure 3.4: Context Diagram (DFD Level 0) of MitraPark

The MitraPark context diagram identifies the flows of information between the system and external entities. Here the external entities are users who visit the system to view their information, users can login logout, upload and manage posts, send and delete messages, manage notifications and admin can delete posts and users.

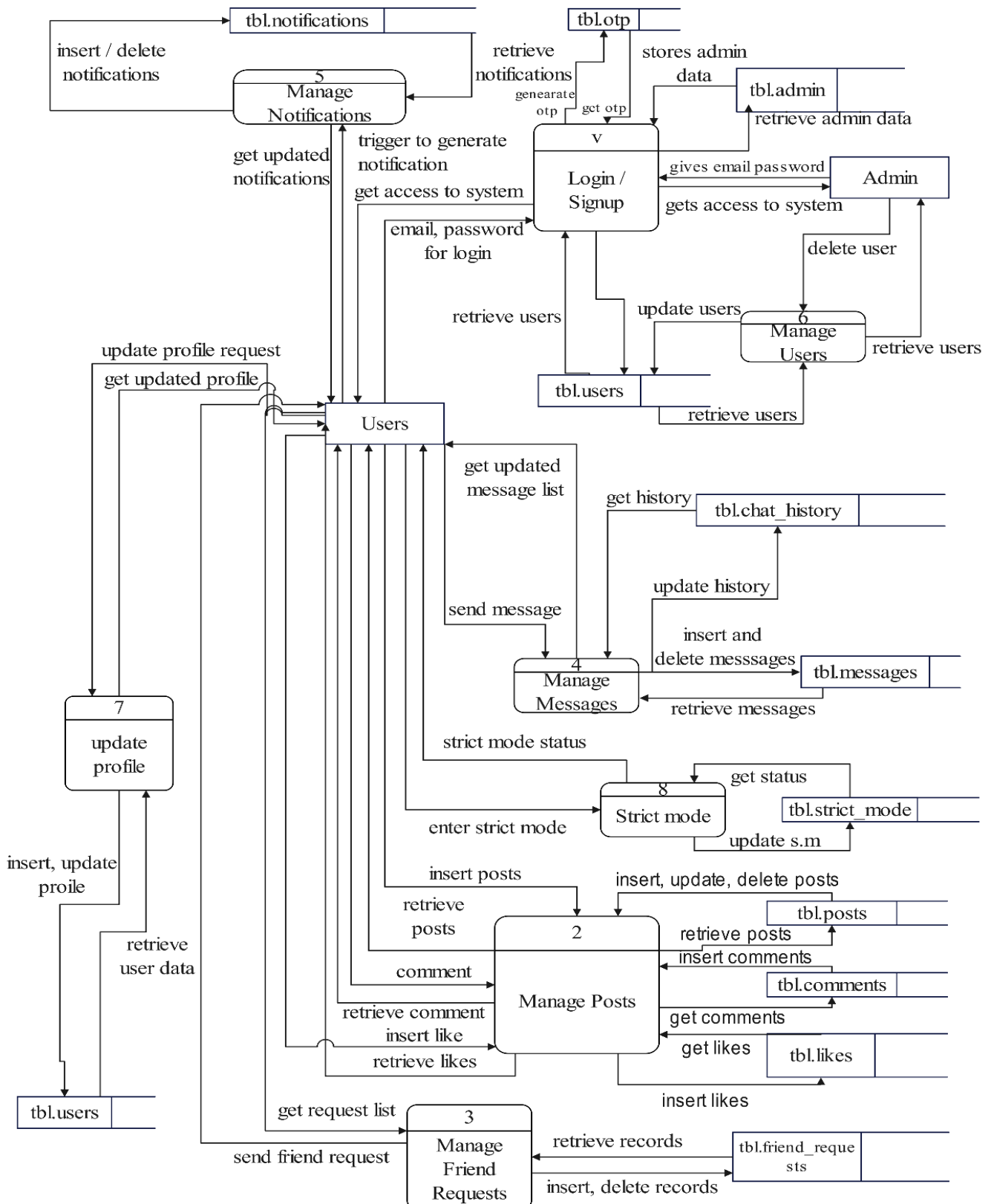


Figure 3.5: DFD Level 1 for MitraPark

In 1-level DFD, the context diagram is decomposed into multiple bubbles/processes. In this level, the main functions of the system were highlighted and breakdown the high-level process of context diagram into sub-processes. The above level 0 DFD shows detailed information about the proposed system. The above figure shows all the actions both users and admins can take.

3.2. System Design

3.2.1. Architecture Design

The system architecture consists of two major parts: the front-end and the back-end, illustrating the basic process flow. The front-end is the interface visible to the user, allowing the use of visual components. It displays different tabs available in the system, such as Feed, Message, Notification, and Settings. For the front-end, HTML, CSS, and JavaScript are utilized, creating an interactive and user-friendly interface. The back-end employs PHP as the server-side scripting language and MySQL as the database, while the Apache Web Server, integrated with XAMPP, is used as the web server. When the user logs into the system, they are redirected to the News Feed, where they can see posts, interact with them using likes and comments, send messages to friends, view notifications, and manage them efficiently.

The following figure shows the architectural design of the Social Media and Social Networking System “MitraPark “

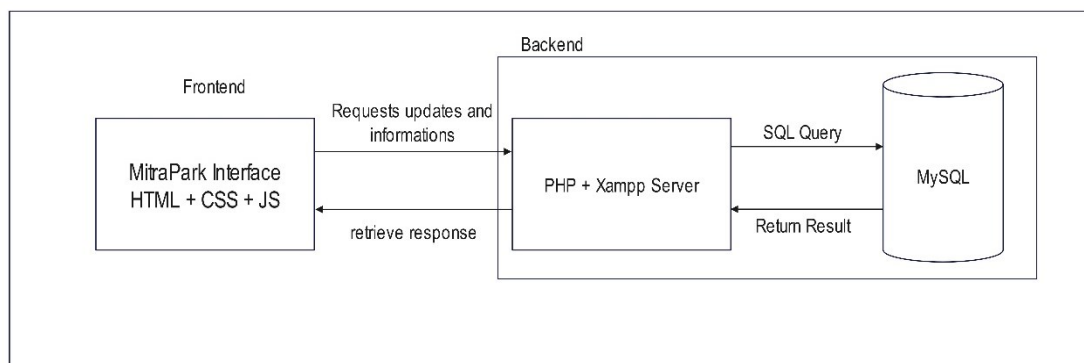


Figure 3.6: Architectural Design of MitraPark

3.2.2. Database Schema Design

The database schema design for Social Media and Social Networking System “MitraPark” showing all the relations along with their attributes and inter-relationships between the relations is shown below:

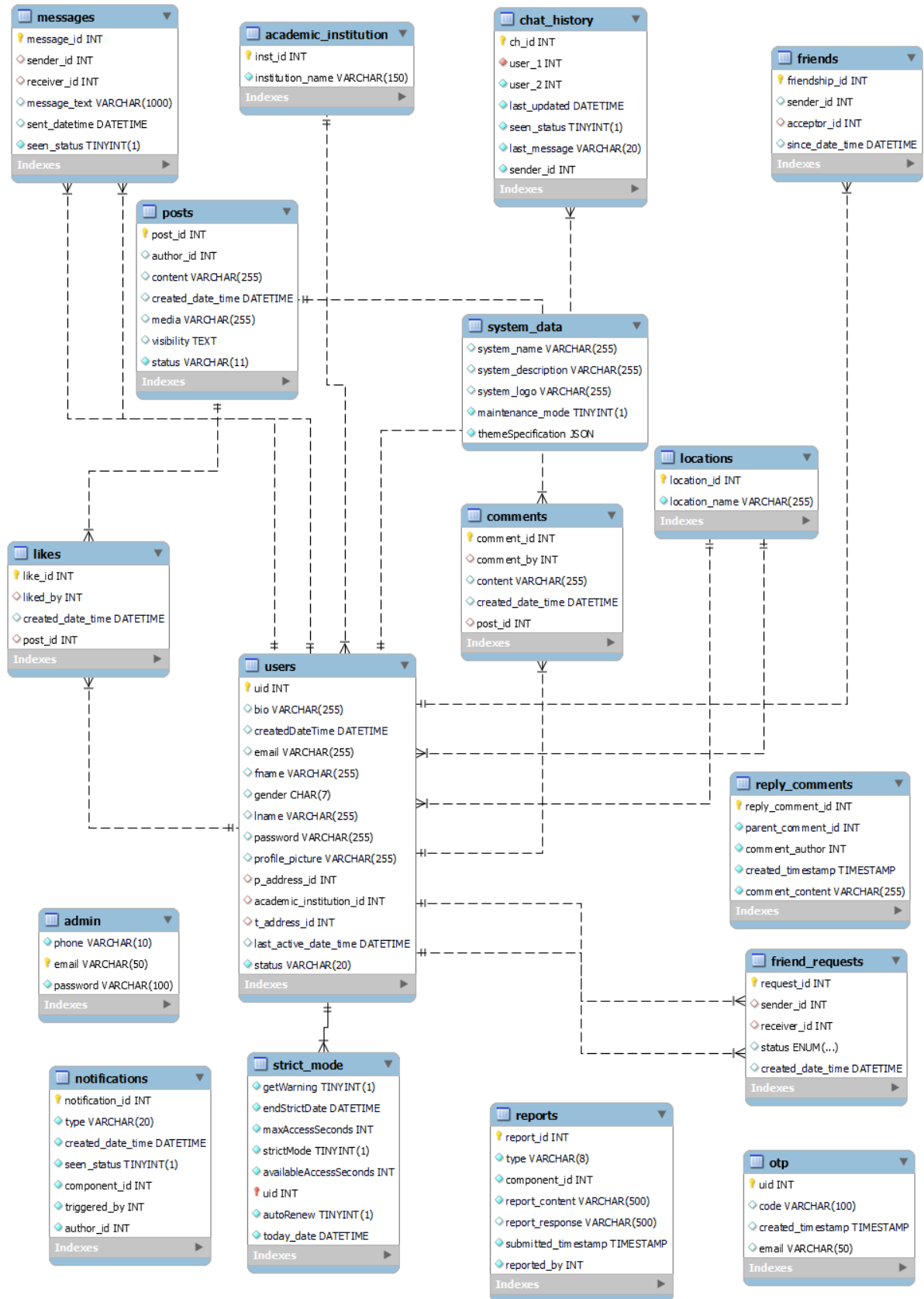


Figure 3.7: Database schema of 'MitraPark'

The database named "MitraPark" has been created for this project, encompassing a comprehensive set of tables to manage various types of information. The `system_data` table holds essential details about the system, including its name, slogan, and logo. User-generated content is managed through several tables: `posts` for storing posts, `likes` for tracking post likes, `comments` for post comments, and `reply_comments` for replies to comments. The `messages` table stores chat data, while `chat_history` keeps a record of past conversations. The `academic_institution` table contains data about academic institutions, and `locations` stores location information. User information is centralized in the `users` table, with `friend_requests` and `friends` tables managing friend interactions and relationships. Notifications are handled by the `notifications` table, and admin credentials are secured in the `admin` table. Additional tables include `otp` for one-time passwords, `reports` for reporting issues, and `strict_mode` for enforcing specific system policies. This well-structured database ensures efficient data management and seamless system functionality.

3.2.3. Interface Design

The interface design (wireframes) has been designed using Adobe XD. It can be used to represent digital sketches of an idea or concept for an application or website, facilitating and understanding before any code is written.

Login Page:

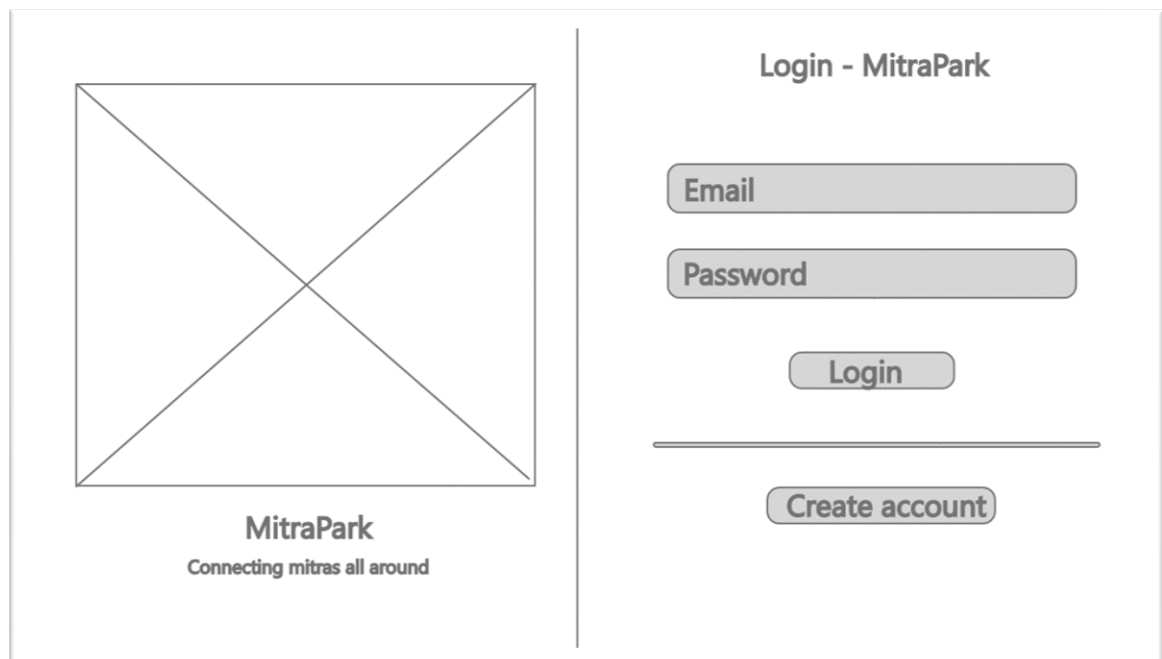


Figure 3.8: Login Page Wireframe

Figure 3.8 shows the login page wireframe. The login page consists of fields entering email, and password. If we have not created an account than we should create an account firstly in login page of the Social Media and Social Networking System 'MitraPark'.

Signup Page:

Signup

First Name Last Name

Email

Phone

Password

Confirm Password

- ⊗ At least one lower case letter
- ⊗ At least one upper case letter
- ⊗ At least one number
- ⊗ At least one special character
- ⊗ Be at least 12 characters

Signup

Figure 3.9: Signup Page Wireframe

Figure 3.9 shows the signup page wireframe. The signup page consists of fields entering First Name, Last Name, Email, Phone, Password, Confirm Password. To fill this fields we have some validation are:

- At least one lower case letter
- At least one upper case letter
- At least one number
- At least one special character
- Be at least 12 characters

Feed Page:



Figure 3.10: Feed Page Wireframe

Figure 3.10 shows the feed Page wireframe. In the feed page, users can add photos, caption to the post. Users can like in the post and comment in the post. On the feed page, users who post and the user's name and users who post the posts, date, and time are also shown.

Mitras Page:

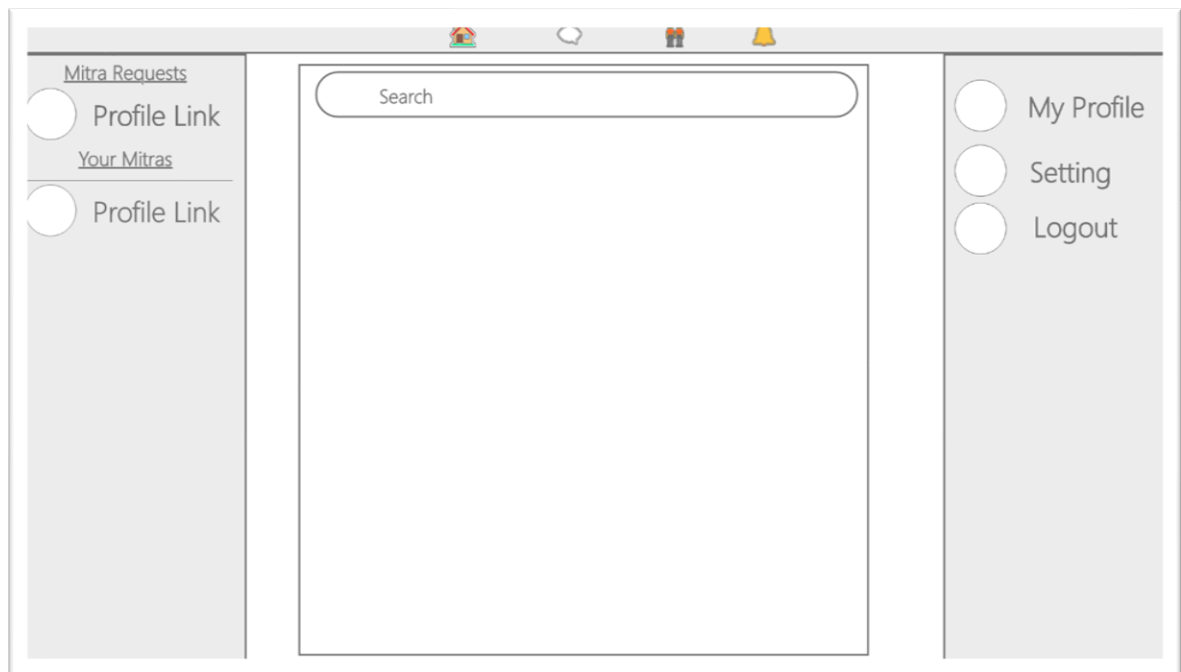


Figure 3.11: Mitras Page Wireframe

Figure 3.11 shows mitras page wireframe. In mitras page we can search for users in MitraPark.

Message Page:

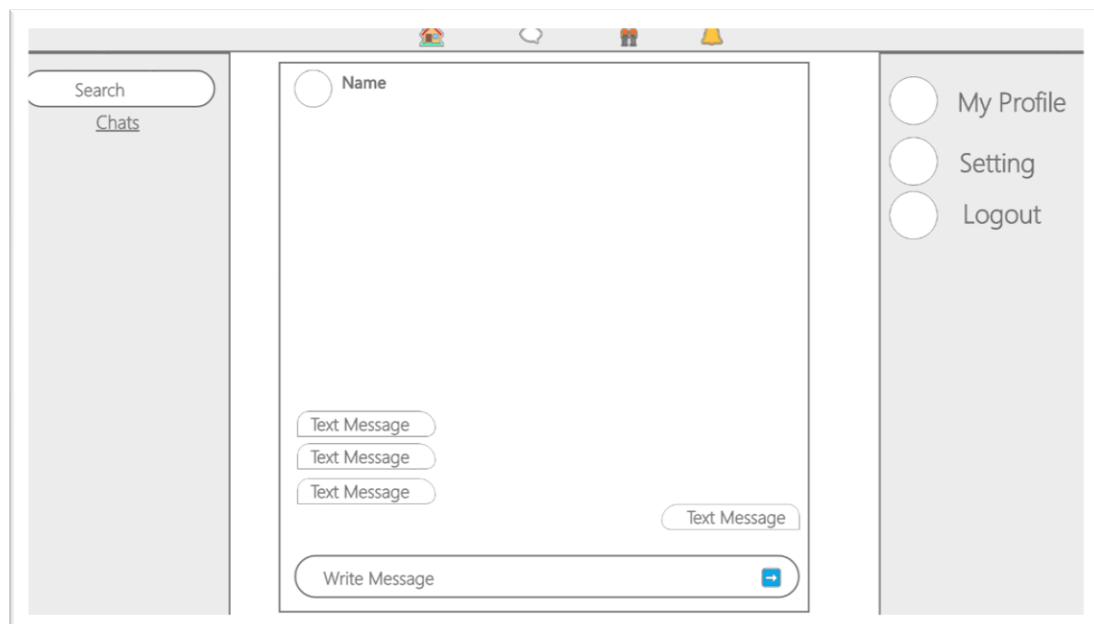


Figure 3.12: Message page wireframe

Figure 3.12 shows the message page wireframe. In message page wireframe we can send

Messages .

Setting Page:

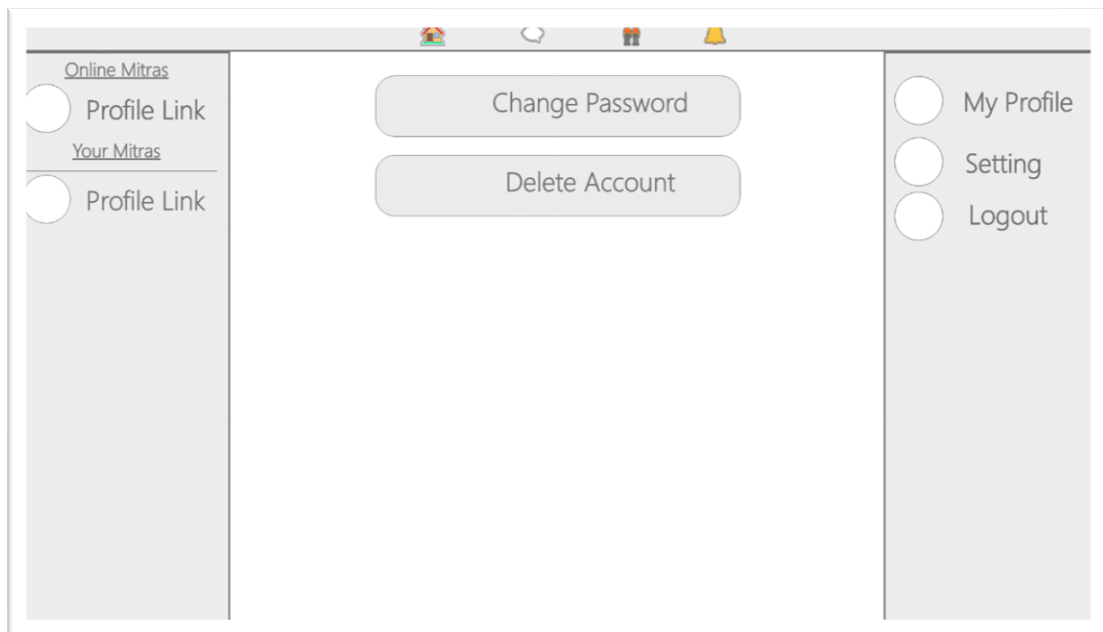


Figure 3.13: Setting Page Wireframe

Figure 3.13 shows setting page wireframe. In setting page wireframe we can change password and delete account .

User Profile Page :

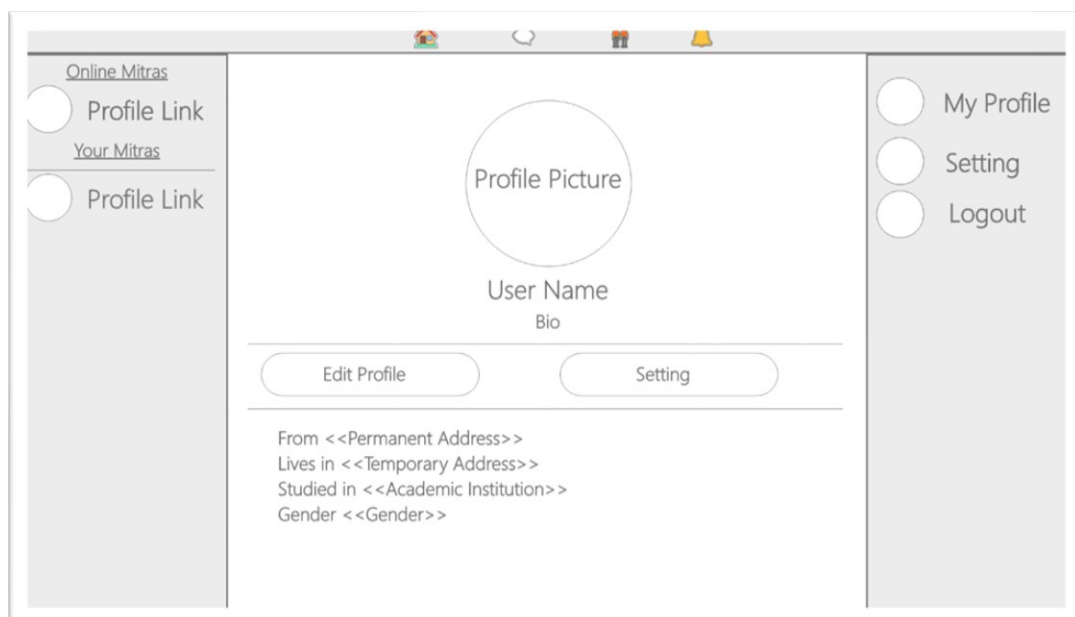


Figure 3.14: Profile page wireframe

Figure 3.14 shows profile page wireframe. Profile page consists of profile picture, user name, bio, edit profile .

Admin Search Post Page:

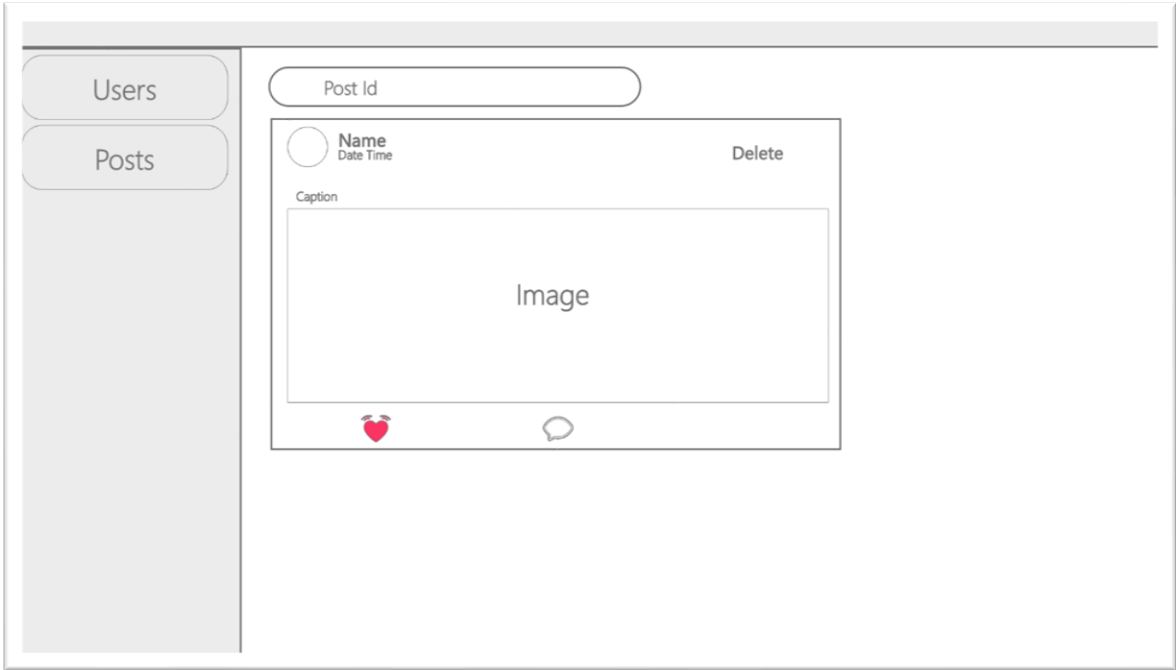


Figure 3.15: Admin Search Post Page Wireframe

Figure 3.15 shows admin search post page wireframe. In this page admin can search the post page .

Notification Page:

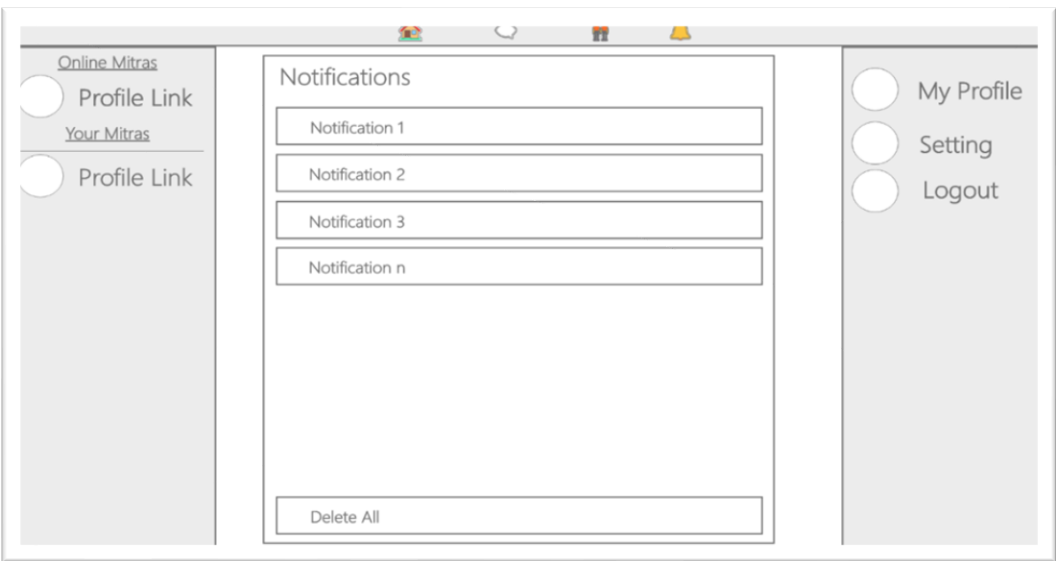


Figure 3.16: Notification page wireframe

Figure 3.16 shows the notification page wireframe. In this page we can see notifications of users who like the post.

Admin search user profiles:

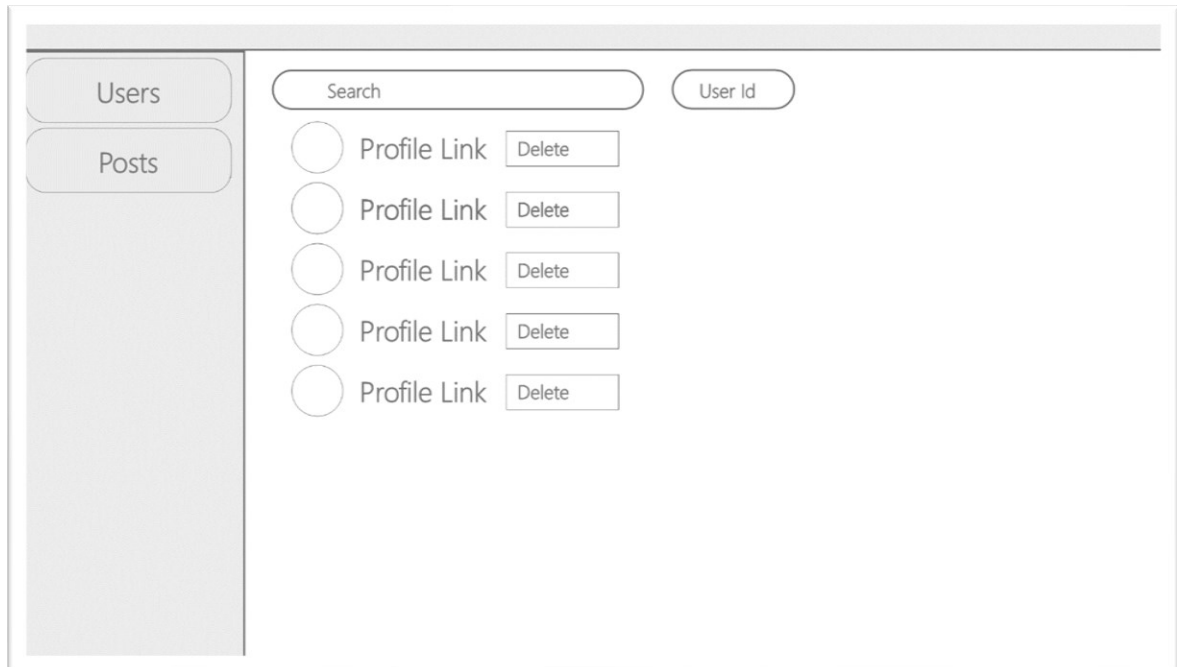


Figure 3.17: Admin Search Profile Page

User Profile Page

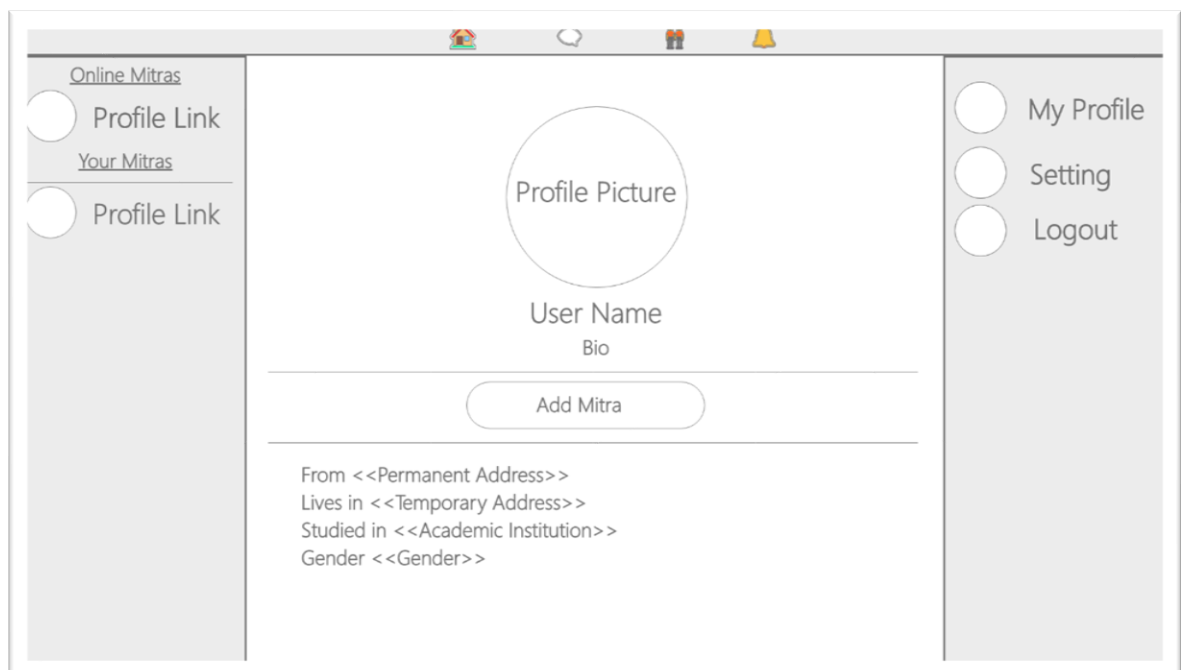


Figure 3.18: User profile wireframe

Admin Dashboard

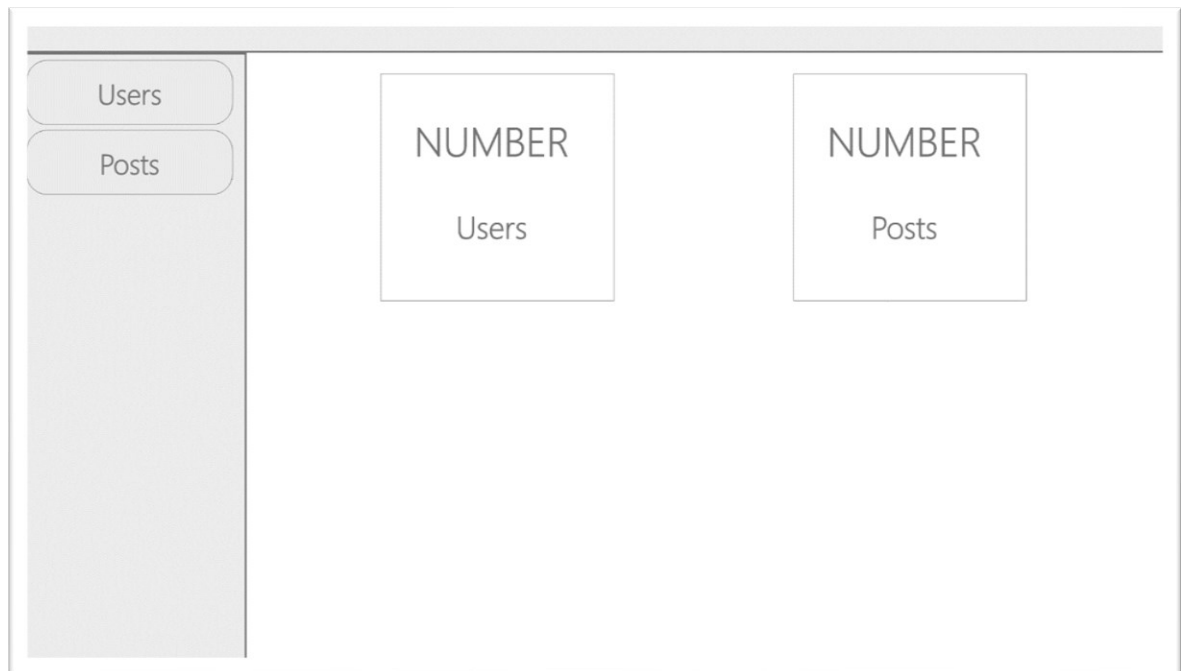


Figure 3.19: Admin Dashboard Page Wireframe

Figure 3.19 shows admin dashboard page wireframe. In this page, there is no of users and no of posts.

3.2.4. Physical DFD

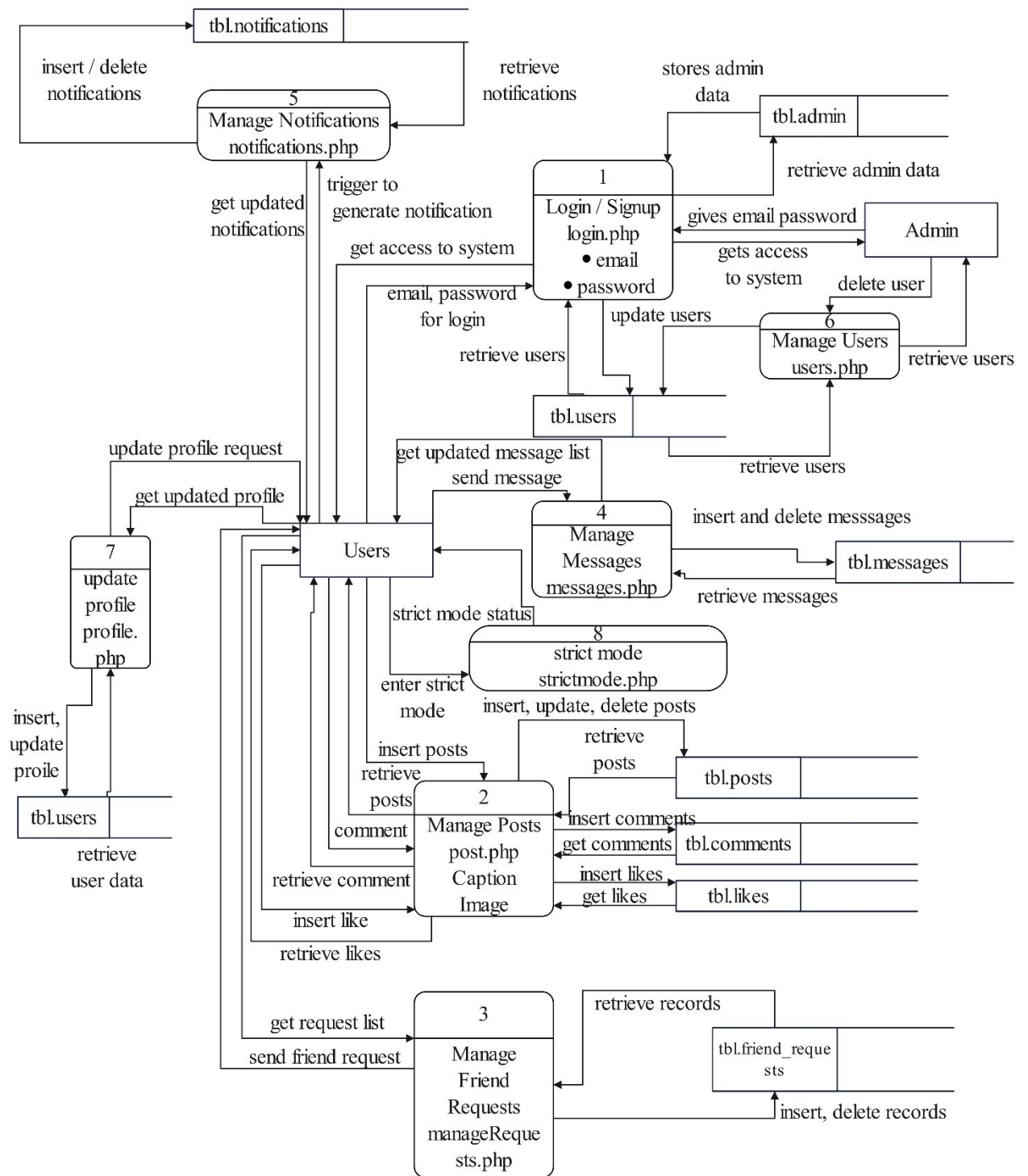


Figure 3. 20: Physical DFD of MitraPark

This system facilitates the exchange of information among various entities involved. The MitraPark system comprises several tables within its database schema, each serving a specific purpose in facilitating interactions and managing data within the platform.

Firstly, the `system_data` table holds critical information about the MitraPark system itself, including its name, slogan, and logo. These details are integral for branding and identification purposes within the application.

Secondly, the `posts` table serves as a repository for user-generated content. Users can create posts containing various types of media, such as text, images, or videos. Each post is associated with the user who authored it and includes details like the content and creation timestamp.

Thirdly, the `likes` table records instances where users express their appreciation for posts by 'liking' them. This table maintains a log of which users liked which posts and when, enabling the system to gauge post popularity and offer tailored recommendations.

The `comments` table functions similarly to likes but focuses on user comments on posts. It stores information about the content of comments, the users who posted them, and the posts they were made on, fostering discussions and feedback among users.

Additionally, the `messages` table captures conversations between users via the MitraPark messaging feature. Each message includes sender and recipient details, message content, and a timestamp, facilitating private communication channels.

The `academic_institution` table contains data about various academic institutions, providing users with information about educational establishments. This data can aid in user networking and connecting with individuals from specific academic backgrounds.

Overall, these tables form the backbone of the MitraPark system, enabling seamless interaction and data management among its diverse user base.

Chapter 4: Implementation and Testing

4.1. Implementation

Implementation is the process through which a decision or plan is translated into effect. During implementation, coding is commenced by the established requirements.

4.1.1 Tools Used Diagram Tool:

“Draw.io” and “Wondershare EdrawMax” is used to make all the system designs required for this project. These are proprietary software for making diagrams and charts. The software lets us choose from an automatic layout function or create a custom layout. The drag-and-drop feature make it simple to create a great looking diagram or chart.

Web Application Development Tools:

Various tools are used to design the web page for this project.

a) HTML:

HTML is a markup language that is used for creating web pages and which defines the structure of web pages. HTML has been employed for the front end, and it stands as one of the fundamental building blocks for every website.

b) CSS:

CSS is the language for describing the presentation of web pages, including colors, layout, and fonts. It is a simple design language intended to simplify the process of making web pages presentable.

c) JavaScript

JS (JavaScript) is a lightweight, interpreted programming language. It is designed for creating network-centric applications. JavaScript is very easy to implement because it is integrated with HTML. It is open and cross-platform.

d) PHP:

PHP is an open-source general-purpose scripting language that is especially suited for web development and can be embedded into HTML. It is a scripting language used to create dynamic websites. A dynamic page has been created with the aid of PHP.

e) MySQL:

MySQL is an open-source relational database management system. It is based on the structure query language, which is used for adding, removing and modifying information in the database. Different types of commands like add, drop, insert, and update can be performed with the assistance of MySQL.

f) Visual Studio Code

Visual Studio Code is a source-code editor that includes features like debugging, syntax highlighting, intelligent code completion, etc. It was made by Microsoft and can run on different types of operating systems like Windows, Linux and Mac OS. Different programming languages are readily available thus making the coding process faster and hassle-free.

g) Apache

Apache is the actual web server application that processes and delivers web content to a computer. Apache is the most popular web server online, powering nearly 54% of all websites.

h) MySQL Workbench

MySQL Workbench is a comprehensive visual tool for database architects, developers, and administrators to design, develop, and manage MySQL databases. It offers features like visual database modeling, SQL development, server administration, and performance monitoring.

4.1.2 Implementation Details of Modules

i. Registration Module

The registration module allows users to input necessary and valid information on the registration form, enabling them to access the system. Once the provided information is verified, users can successfully register and gain access to the system.

ii. Login Module

The login module grants access to the system for users. Upon successful verification, users can log in to their respective accounts and access the system's functionalities.

iii. Logout Module

The logout module provides users with a secure way to end their sessions and log out from the system. When users initiate the logout process, the system terminates their active session, ensuring that their access to system features and data is revoked.

iv. Add Post Module

A logged-in user can upload a post with some text caption and image file.

v. Delete Post Module

When the user wants to delete a post published by them it is done through this module.

vi. Edit Post Module

When the user has to update (make changes) in previously uploaded post's caption or photo it is done through this module.

vii. Get Posts Module

When a user is logged in they are redirected to news feed this module is implemented in that page where it provides posts published as Visibility Public, Posts Published by Mitras and own private posts.

viii. Send Messages Module

In the chat section when a user has to chat with other user this module takes text message from the user and sends to the selected user.

ix. Get Messages Module

In the chat section when a user clicks on chat this module retrieves previous chats done and displays it.

x. Delete Message Module

This module allows users to delete a message sent by them self permanently .

xi. Add Like Module

When a user clicks on like button in a post this module increases the like count and updates the likers list of the post.

xii. Remove Like Module

When a user clicks on like button where the state of like button is already liked then the like is removed.

xiii. Add Comment Module

The user can add comments with some text in a post using this module.

xiv. Delete Comment Module

When the user wants to delete a comment published by them it is done through this module.

xv. Edit Comment Module

The edit comment module allows users to update or make changes to previously published comments. Users can modify the text of their comments and resubmit it. This module provides an editable field that displays the current content of the comment. Upon submission, the changes are saved to the database, and the updated comment is displayed under the post.

xvi. Get Comment Module

The get comment module retrieves and displays comments on a post. When a user clicks the comment button, this module fetches all comments related to the post from the database and displays them in chronological order. This module ensures that comments are loaded efficiently and includes features such as displaying the commenter's name and profile picture, as well as the comment's timestamp.

xvii. Search User Module

The search user module allows users to search for other users by their name. When a user enters a name in the search field, this module queries the database for matching user profiles and displays the results. It ensures that the search operation is performed quickly and accurately.

xviii. Send Friend Request (Add Mitra) Module

The send friend request module enables users to send friend requests to other users. When a user finds another user they want to connect with, they can send a friend request through this module. The request is saved in the database and a notification is sent to the recipient. This module ensures that users can easily initiate connections and includes features such as confirmation messages and the ability to cancel sent requests.

xix. Accept Friend Request (Add Mitra) Module

The accept friend request module allows users to accept friend requests sent to them. When a user receives a friend request, they can view the request and choose to accept it. Upon acceptance, the users are added to each other's friend lists, and the relationship is saved in the database. This module ensures that new connections are established smoothly and includes features such as notifications of new friends and the ability to view the sender's profile before accepting.

xx. Remove Friend Module

The remove friend module enables users to remove someone from their friend list. When a user decides to unfriend another user, they can do so through this module. This action removes the relationship from the database and updates the friend lists.

xxi. Change Password Module

The change password module allows users to update their existing password. When a user needs to change their password, they can enter their current password and the new desired password into the provided fields. This module verifies the current password and updates the user's credentials in the database. It includes features such as password strength indicators and validation rules to ensure that the new password meets security standards.

xxii. Add strict mode Module

If user has to activate strict mode this module is used to enable strict mode.

xxiii. Remove strict mode Module

If user has to remove strict mode this module is used to disable strict mode.

xxiv. Report user Module

A user can report another user using this module.

xxv. Restrict reported user Module

Admin can restrict reported users using this module.

xxvi. Report post Module

A user can report any post using this module.

xxvii. Restrict reported Posts

Admin can restrict a reported post using this module.

xxviii. Unrestrict restricted user Module

Admin can unrestrict users using this module.

xxix. Unrestrict restricted post Module

Admin can unrestrict post using this module.

4.2 Testing

Testing is a systematic process of evaluating a software application or system to identify any defects, errors, or discrepancies between expected and actual behavior. It involves executing the software with various test cases and scenarios to verify its functionality, performance, reliability, and security. The goal of testing is to ensure that the software meets its requirements, performs as expected, and delivers a high-quality user experience. Testing helps to uncover bugs and issues early in the development lifecycle, allowing them to be addressed before the software is deployed to users.

4.2.1. Test Cases for Unit Testing

Each view/module of the application is tested individually. As the modules were built up testing was carried out simultaneously, tracking out each and every kind of input and checking the corresponding output until module is working correctly.

Table 4. 1 Test case for user Login / Signup

Project Name : Social Media and Social Networking System “MitraPark”	
Test Case : For Valid Login Attempt	
Test Case ID: TC_1	Test Designed by: Alen Pariyar
Test Priority(Low/Medium/High): High	Test Designed Date: 2024-01-02
Module Name: Login Form Credentials	Test Executed by: Alen Pariyar
Test title: Verify login validation	Test Execution date: 2024-01-05
Description: Test the MitraPark’s login page	
Pre-condition: User has valid email and password	

Step	Test Steps	Test Data	Expected Result	Pass / Fail
1	Navigate to the login page		The login page should open.	Pass
2	Input valid email	oct10.alenpariyar@gmail.com	The username should be entered	Pass
3	Input valid password	9816699413##Alen	The password should be entered	Pass
4	Click on the Login Button		The login button should be clickable, After click it should be redirected to feed page.	Pass
Post Conditions : Login credentials are validated with credentials in database and successfully logged into System.				

Table 4. 2 Test case for Invalid user login

Project Name : Social Media and Social Networking System “MitraPark”	
Test Case : For Invalid Login Attempt	
Test Case ID: TC_2	Test Designed by: Alen Pariyar

Test Priority(Low/Medium/High): High		Test Designed Date: 2024-01-02		
Module Name: Login Form		Test Executed by: Alen Pariyar		
Test title: Verify login validation Credentials		Test Execution date: 2024-01-05		
Description: Test the MitraPark’s login page				
Pre-condition: Invalid email and password should be entered				
Step	Test Steps	Test Data	Expected Result	Pass / Fail
1	Navigate to the login page		The login page should open.	Pass
2	Input valid email	alenpariyar@gmail.com	The username should be entered	Pass
3	Input valid password	9816699413@@Alen	The password should be entered	Pass
4	Click on the Login Button		The login button should be clickable, After click it should show ‘Couldn’t find user.’	Pass

Post Conditions : Login credentials are validated with credentials in database and redirected to login page with message “Couldn’t find user.”

Table 4. 3 Test case for user login form validation

Project Name : Social Media and Social Networking System “MitraPark”					
Test Case : Login Form Validation					
Test Case ID: TC_3			Test Designed by: Alen Pariyar		
Test Priority(Low/Medium/High): High			Test Designed Date: 2024-01-02		
Module Name: Login Form			Test Executed by: Alen Pariyar		
Test title: Verify login validation			Test Execution date: 2024-01-05		
Description: Test the MitraPark’s login page					
Pre-condition: Invalid email and password should be entered					
Step	Test Scenario	Test Steps	Test Data	Expected Result	Pass / Fail
1		Navigate to the login page		The login page should open.	Pass
2	A valid Email must contain : A valid email formatted string	Input valid email	alenpariyar#rmail.c	The username should be entered	Pass
3	A valid password must contain: At least 12 characters and at most 18 characters.	Input valid password	98166alen	The password should be entered	Pass
4	Since, the input values are not following the email and password rules. The login button	Click on the Login Button		The login button should not be clickable and stay disabled.	Fail

	should be in disabled state.				
Post Conditions : Due to form validation and invalid inputs the form cannot be submitted.					

Table 4. 4 Test case for User Signup

Project Name : Social Media and Social Networking System “MitraPark”					
Test Case : Signup Form Validation					
Test Case ID: TC_4			Test Designed by: Alen Pariyar		
Test Priority(Low/Medium/High): High			Test Designed Date: 2024-01-05		
Module Name: Signup Form			Test Executed by: Alen Pariyar		
Test title: Verify s i g n u p validation			Test Execution date: 2024-01-06		
Description: Test the MitraPark’s signup page					
Pre-condition: The user should have all the necessary details, Valid Email, Valid Names The password field must have: • At least one lowercase letter, • At least one uppercase letter, • At least one number, • At least one special character, • Be at least 12 characters and at most 18 characters • Password and Confirm password must match					
Step	Test Scenario	Test Steps	Test Data	Expected Result	Pass / Fail
1		Navigate to the signup page		The Signup page should open.	Pass

2	A valid first name : Must contain only the alphabet, with at least 2 characters and at most 10 characters.	Enter valid first name	Alen	The first name must be entered and no error messages should appear	Pass
3	A valid last name : Must contain only the alphabet, with at least 2 characters and at most 10 characters.	Input valid last name	Pariyar	The last name must be entered and no error messages should appear	Pass
4	A valid password that follows the rules mentioned above.	Input valid password	9816699413# #Alen	The password must be entered and no error messages should appear.	Pass
5	A valid confirm password matching the password	Input valid confirm password	9816699413# #Alen	The confirm password must be entered and no error messages should appear.	Pass
6	After entering all valid credentials the Signup button will be clickable.	Click on Signup button		After clicking the page must redirect to Email verification page.	Pass

7	A 6 digit OTP (One-time password) is sent in valid email address	Enter the OTP code received in the mentioned email address.	317671	Redirect to login page with message “Account created successfully. Please login to continue”	Pass
Post Conditions : In all valid case scenarios the signup will be completed and user account is created.					

Table 4. 5 Test case for Signup with invalid OTP code

Project Name : Social Media and Social Networking System “MitraPark”					
Test Case : Signup Form Validation – Using Invalid OTP					
Test Case ID: TC_5			Test Designed by: Alen Pariyar		
Test Priority(Low/Medium/High): High			Test Designed Date: 2024-01-05		
Module Name: Signup Form			Test Executed by: Alen Pariyar		
Test title: Verify s i g n u p validation			Test Execution date: 2024-01-06		
Description: Test the MitraPark’s signup page					
Pre-condition: Using wrong OTP while signup process.					
Step	Test Scenario	Test Steps	Test Data	Expected Result	Pass / Fail
1		Navigate to the signup page		The Signup page should open.	Pass
2	A valid first name : Must contain only the alphabet, with at least 2 characters and at most 10 characters.	Enter valid first name	Alen	The first name must be entered and no error messages should appear	Pass

3	A valid last name : Must contain only the alphabet, with at least 2 characters and at most 10 characters.	Input valid last name	Pariyar	The last name must be entered and no error messages should appear	Pass
4	A valid password that follows the rules mentioned above.	Input valid password	9816699413# #Alen	The password must be entered and no error messages should appear.	Pass
5	A valid confirm password matching the password	Input valid confirm password	9816699413# #Alen	The confirm password must be entered and no error messages should appear.	Pass
6	After entering all valid credentials the Signup button will be clickable.	Click on Signup button		After clicking the page must redirect to Email verification page.	Pass
7	A 6 digit OTP (One-time password) is sent in valid email address	Enter the wrong OTP code.	123456	Should show message "Invalid OTP"	Pass
Post Conditions : Use of invalid OTP code the signup process cannot be completed.					

Table 4. 6 Test case for signup with existing user credentials

Project Name : Social Media and Social Networking System “MitraPark”					
Test Case : Signup Form Validation – Using existing user credentials					
Test Case ID: TC_6			Test Designed by: Alen Pariyar		
Test Priority(Low/Medium/High): High			Test Designed Date: 2024-01-05		
Module Name: Signup Form			Test Executed by: Alen Pariyar		
Test title: Verify s i g n u p validation			Test Execution date: 2024-01-06		
Description: Test MitraPark’s signup OTP code					
Pre-condition: Using the email address in signup form which has already been used in existing user account.					
Step	Test Scenario	Test Steps	Test Data	Expected Result	Pass / Fail
1		Navigate to the signup page		The Signup page should open.	Pass
2	A valid first name : Must contain only the alphabet, with at least 2 characters and at most 10 characters.	Enter valid first name	Alen	The first name must be entered and no error messages should appear	Pass
3	A valid last name : Must contain only the alphabet, with at least 2 characters and at most 10 characters.	Input valid last name	Pariyar	The last name must be entered and no error messages should appear	Pass

4	A valid password that follows the rules mentioned above.	Input valid password	9816699413# #Alen	The password must be entered and no error messages should appear.	Pass
5	A valid confirm password matching the password	Input valid confirm password	9816699413# #Alen	The confirm password must be entered and no error messages should appear.	Pass
6	After entering all valid credentials the Signup button will be clickable.	Click on Signup button		After clicking the page must redirect to Email verification page.	Pass
7				It must show “User already exists” message	Pass
Post Conditions : Use of existing account email the signup process cannot be completed.					

Table 4. 7 Test case for invalid OTP in password reset

Project Name : Social Media and Social Networking System “MitraPark”	
Test Case : Forgot password	
Test Case ID: TC_7	Test Designed by: Alen Pariyar
Test Priority(Low/Medium/High): High	Test Designed Date: 2024-01-05
Module Name: Forgot password	Test Executed by: Alen Pariyar

Test title: Verify forgot password			Test Execution date: 2024-01-06		
Description: Test the Forgot password module					
Pre-condition: Using wrong OTP in forgot password operation					
Step	Test Scenario	Test Steps	Test Data	Expected Result	Pass / Fail
1		Navigate to the signup page		The Signup page should open.	Pass
2	A valid email, existing user account	Enter valid email address	oct10.alenpariyar@gmail.com	Send button should be clickable	Pass
3	After entering the valid email address	Click on Send button		It should be redirected to Verify OTP and OTP should be sent to Provided mail address.	Pass
4	A 6 digit OTP (One-time password) is sent in valid email address	Enter the wrong OTP code.	123456	Should show message “Invalid OTP”	Fail
Post Conditions : Use of invalid OTP code the password reset process cannot be completed.					

Table 4. 8 Test case reset password

Project Name : Social Media and Social Networking System "MitraPark"	
Test Case : Forgot password	
Test Case ID: TC_8	Test Designed by: Alen Pariyar
Test Priority(Low/Medium/High): High	Test Designed Date: 2024-01-06

Module Name: Forgot password			Test Executed by: Alen Pariyar		
Test title: Verify forgot password			Test Execution date: 2024-01-07		
Description: Test the Forgot password module					
Pre-condition: Using OTP received in Email to reset password					
Step	Test Scenario	Test Steps	Test Data	Expected Result	Pass / Fail
1		Navigate to the signup page		The Signup page should open.	Pass
2	A valid email, existing user account	Enter valid email address	oct10.alenpari yar@gmail.co m	Send button should be clickable	Pass
3	After entering the valid email address	Click on Send button		It should be redirected to Verify OTP and OTP should be sent to Provided mail address.	Pass
4	A 6 digit OTP (One-time password) is sent in valid email address	Enter the OTP code sent in email address.	592517	Redirect to change password	Pass
5	A valid password that matches the provided rules.	Enter valid new password.	9823435146 @#Alen	The password must be entered and no error messages related to password must appear.	Pass

6	A valid confirm password that matches the password.	Repeat the password value.	9823435146 @#Alen	The password must be entered and no error messages related to password must appear.	Pass
7	After clicking “Change Password”			Show message “Password changed successfully”	Pass
Post Conditions : Changing password can be done using valid email and OTP sent in email.					

Table 4. 9 Test case to upload post

Project Name : Social Media and Social Networking System “MitraPark”					
Test Case : Upload a post					
Test Case ID: TC_9			Test Designed by: Alen Pariyar		
Test Priority(Low/Medium/High): High			Test Designed Date: 2024-01-07		
Module Name: Upload Post			Test Executed by: Alen Pariyar		
Test title: Uploading a post			Test Execution date: 2024-01-07		
Description: Test to upload a post					
Pre-condition: Must be logged in the system					
Step	Test Scenario	Test Steps	Test Data	Expected Result	Pass / Fail
1		Navigate to the feed page		The Signup page should open.	Pass

2		Click on “Write something” box.		A “create post” popup will appear,	Pass
3		The post must contain (both or any one) • Caption • Photo Select a visibility mode and Click share	Caption , Photo, Visibility Mode	Post should be uploaded and appear in feed page	Pass
4	A 6 digit OTP (One-time password) is sent in valid email address	Enter the OTP code sent in email address.	592517	Redirect to change password	Pass
Post Conditions : Using valid post caption and image post can be uploaded					

Table 4. 10 Test case upload post without caption and image

Project Name : Social Media and Social Networking System “MitraPark”					
Test Case : Upload a post – invalid attempt					
Test Case ID: TC_10			Test Designed by: Alen Pariyar		
Test Priority(Low/Medium/High): High			Test Designed Date: 2024-01-07		
Module Name: Upload Post			Test Executed by: Alen Pariyar		
Test title: Uploading a post			Test Execution date: 2024-01-07		
Description: Test to upload a post without caption and phone					
Pre-condition: Must be logged in the system					
Step	Test Scenario	Test Steps	Test Data	Expected Result	Pass / Fail

1		Navigate to the feed page		The Signup page should open.	Pass
2		Click on the “Write something” box.		A “create post” popup will appear,	Pass
3	The post must contain (both or anyone) • Caption • Photo	Not providing any of • Caption • Photo	Caption, Photo, Visibility Mode	The share button should be disabled and the form must not be submitted.	Pass
Post Conditions: Using valid post captions and image posts can be uploaded					

Table 4. 11 Test case editing post

Project Name: Social Media and Social Networking System “MitraPark”					
Test Case: Editing a post					
Test Case ID: TC_11			Test Designed by: Alen Pariyar		
Test Priority(Low/Medium/High): High			Test Designed Date: 2024-01-08		
Module Name: Edit Post			Test Executed by: Alen Pariyar		
Test title: Editing a post			Test Execution date: 2024-01-08		
Description: Test to edit an uploaded post					
Pre-condition: Must be logged in the system, Must have posted a post before					
Step	Test Scenario	Test Steps	Test Data	Expected Result	Pass / Fail
1		Navigate to the post uploaded before		The post single page must appear	Pass

2		Click on “Edit Post”		A “edit post” popup will appear,	Pass
3		Edit any of or all of : Caption, Visibility, Photo	Caption, Photo, Visibility Mode	The save button must be clickable	Pass
4		Click on save button		Form must be submitted and post changes should be reflected	Pass
Post Conditions : Using valid post caption and image post can be edited					

Table 4. 12 Test case for deleting a post

Project Name: Social Media and Social Networking System “MitraPark”					
Test Case: Deleting a post					
Test Case ID: TC_12			Test Designed by: Alen Pariyar		
Test Priority(Low/Medium/High): High			Test Designed Date: 2024-01-09		
Module Name: Delete Post			Test Executed by: Alen Pariyar		
Test title: Deleting a post			Test Execution date: 2024-01-09		
Description: Test to delete an uploaded post					
Pre-condition: Must be logged in the system, Must have posted a post before					
Step	Test Scenario	Test Steps	Test Data	Expected Result	Pass / Fail
1		Navigate to the post uploaded before		The post's single page must appear	Pass

2		Click on “Delete Post”		A “delete post” confirm box will appear,	Pass
3		Click on “Yes”		The post will be deleted from the system	Pass
4		Click on the save button		Form must be submitted and post changes should be reflected	Pass
Post Conditions: The post will not be further available in the system.					

Table 4. 13 Test case to add comment

Project Name: Social Media and Social Networking System “MitraPark”					
Test Case: Adding comment in a post					
Test Case ID: TC_13			Test Designed by: Alen Pariyar		
Test Priority(Low/Medium/High): Medium			Test Designed Date: 2024-01-09		
Module Name: Add comment			Test Executed by: Alen Pariyar		
Test title: Adding comment in the post			Test Execution date: 2024-01-09		
Description: Test to add a comment in an uploaded post					
Pre-condition: Must be logged in the system, Must have posted a post before					
Step	Test Scenario	Test Steps	Test Data	Expected Result	Pass / Fail
1		Navigate to the post uploaded before		The post's single page must appear	Pass

2		Click on “Write comment”		The cursor will be blinking in the comment textbox.	Pass
3		Type a comment and click on the send icon.	hey	The comment will appear in the post.	Pass
Post Conditions: The comment can be seen in the comment section of the post. The notification is sent to the author of the post.					

Table 4. 14 Test case to edit a comment

Project Name: Social Media and Social Networking System “MitraPark”					
Test Case: Editing comment in a post					
Test Case ID: TC_14			Test Designed by: Alen Pariyar		
Test Priority(Low/Medium/High): Medium			Test Designed Date: 2024-01-10		
Module Name: Edit comment			Test Executed by: Alen Pariyar		
Test title: Editing comment in the post			Test Execution date: 2024-01-10		
Description: Test to edit a existing comment in an uploaded post					
Pre-condition: Must be logged in the system, Must have posted a post before, Must commented before					
Step	Test Scenario	Test Steps	Test Data	Expected Result	Pass / Fail
1		Navigate to the post uploaded before		The post's single page must appear	Pass
2		Click on the “Comments” section		Lists of comments will appear.	Pass

3		Click on the edit comment		A textbox with the current text in a comment should appear.	Pass
4		Change the comment content and click on save	hello	The comment should be updated.	Pass
Post Conditions: The comment can be seen in the comment section of the post.					

Table 4. 15 Test case to delete a comment

Project Name: Social Media and Social Networking System “MitraPark”					
Test Case: Deleting a comment in a post					
Test Case ID: TC_15			Test Designed by: Alen Pariyar		
Test Priority(Low/Medium/High): Medium			Test Designed Date: 2024-01-10		
Module Name: Delete comment			Test Executed by: Alen Pariyar		
Test title: Deleting comment in the post			Test Execution date: 2024-01-10		
Description: Test to delete an existing comment in an uploaded post					
Pre-condition: Must be logged in the system, Must have posted a post before, Must commented before					
Step	Test Scenario	Test Steps	Test Data	Expected Result	Pass / Fail
1		Navigate to the post uploaded before		The post's single page must appear	Pass
2		Click on the “Comments” section		Lists of comments should appear.	Pass

3		Click on the “Confirm Delete”		The comment must be deleted.	Pass
Post Conditions: The comment cannot be further seen in the comment section of the post.					

Table 4. 16 Test case to like a post

Project Name: Social Media and Social Networking System “MitraPark”					
Test Case: Like in a post					
Test Case ID: TC_16			Test Designed by: Alen Pariyar		
Test Priority(Low/Medium/High): Medium			Test Designed Date: 2024-01-10		
Module Name: Add like			Test Executed by: Alen Pariyar		
Test title: L i k e in the post			Test Execution date: 2024-01-10		
Description: Test to add a like in an existing uploaded post					
Pre-condition: Must be logged in the system, Must have available posts in feed page					
Step	Test Scenario	Test Steps	Test Data	Expected Result	Pass / Fail
1		Navigate to the feed page		The list of posts must appear	Pass
2		Click on the ‘love’ icon		The shape should be changed from outline to fill and like count must be incremented	Pass
Post Conditions: The like can be seen in the post and like author list can be seen in the “Likes” section. The notification is sent to author of the post.					

Table 4. 17 Test case to remove a like

Project Name: Social Media and Social Networking System “MitraPark”					
Test Case: unlike in a post					
Test Case ID: TC_17			Test Designed by: Alen Pariyar		
Test Priority(Low/Medium/High): Medium			Test Designed Date: 2024-01-10		
Module Name: add like			Test Executed by: Alen Pariyar		
Test title: removing like from the post			Test Execution date: 2024-01-10		
Description: Test to remove a like from liked post					
Pre-condition: Must be logged in the system, Must have available posts in feed page, Must have liked a post					
Step	Test Scenario	Test Steps	Test Data	Expected Result	Pass / Fail
1		Navigate to the feed page		The list of posts must appear	Pass
2		Click on the ‘love’ icon		The shape should be changed from fill to outline and like count must be decremented	Pass
Post Conditions: The like must be removed from the post.					

Table 4. 18 Test case reply to a comment

Project Name: Social Media and Social Networking System “MitraPark”	
Test Case: Replying comment in a post comment	
Test Case ID: TC_18	Test Designed by: Alen Pariyar
Test Priority(Low/Medium/High): Medium	Test Designed Date: 2024-01-09

Module Name: insertReplyComment			Test Executed by: Alen Pariyar		
Test title: replying to a comment in the post			Test Execution date: 2024-01-09		
Description: Test to reply a comment in an uploaded post					
Pre-condition: Must be logged in the system, Must have posted a post before, Must have comments in the post					
Step	Test Scenario	Test Steps	Test Data	Expected Result	Pass / Fail
1		Navigate to the post uploaded before		The post's single page must appear	Pass
2		Click on the “Comments” section		The list of comments should appear.	Pass
3		Click on the “Reply Comment” button		A popup must appear with the comment box.	Pass
4		Type a comment in the textbox and click on submit	Test comment	A popup of success must appear	Fail
Post Conditions: The replied comment can be seen in the comment section of the post.					

Table 4. 19 Test case editing reply comment

Project Name: Social Media and Social Networking System “MitraPark”	
Test Case: Editing replied comment in a post	
Test Case ID: TC_19	Test Designed by: Alen Pariyar
Test Priority(Low/Medium/High): Medium	Test Designed Date: 2024-01-10
Module Name: updateReplyComment	Test Executed by: Alen Pariyar

Test title: Editing replied comment in the post			Test Execution date: 2024-01-10		
Description: Test to edit an existing replied comment in an uploaded post					
Pre-condition: Must be logged in the system, Must have posted a post before, Must replied in a comment before					
Step	Test Scenario	Test Steps	Test Data	Expected Result	Pass / Fail
1		Navigate to the post uploaded before		The post's single page must appear	Pass
2		Click on the “Comments” section		The list of comments should appear.	Pass
3		Click on the “Edit”		A text box must appear	Pass
4		Type the updated comment in the textbox and click on save	Test reply comment	The replied comment will be updated.	Pass
Post Conditions: The comment can be seen in the comment section of the post.					

Table 4. 20 Test case deleting a reply comment

Project Name: Social Media and Social Networking System “MitraPark”	
Test Case: Deleting a replied comment in a post	
Test Case ID: TC_20	Test Designed by: Alen Pariyar
Test Priority(Low/Medium/High): Medium	Test Designed Date: 2024-01-10
Module Name: deleteReplyComment	Test Executed by: Alen Pariyar
Test title: Deleting replied comment in the post	Test Execution date: 2024-01-10

Description: Test to delete an existing comment in an uploaded post					
Pre-condition: Must be logged in the system, Must have posted a post before, Must replied in a comment before					
Step	Test Scenario	Test Steps	Test Data	Expected Result	Pass / Fail
1		Navigate to the post uploaded before		The post's single page must appear	Pass
2		Click on the "Comments" section		Lists of comments and replied comments should appear.	Pass
3		Click on the "Confirm Delete" in a replied comment		The comment must be deleted.	Pass
Post Conditions: The comment cannot be further seen in the comment section of the post.					

Chapter 5: Conclusion and Future Recommendations

5.1 Lesson Learnt/ Outcome

Upon the conclusion of our project titled "Social Media and Social Networking", our understanding of project development and management has been enriched with a multitude of lessons that have been fortified. Throughout the course of this endeavor, the intricate workings of designing a functional system were not only grasped but also abilities in teamwork and efficient resource management were honed.

The utilization of modern technologies, such as HTML, CSS, PHP, and MySQL, to create a user-friendly and efficient platform, is one of the key takeaways from this project.

A pivotal role was played by collaboration in the successful execution of this project. As a team, the importance of effective communication, division of tasks, and mutual support was discovered. These experiences have fortified the ability to navigate collaborative projects in the future with heightened efficiency and cohesion.

5.2. Conclusion

MitraPark has successfully integrated a range of features and functionalities that significantly enhance its role in community engagement. The meticulous implementation of user registration and validation processes ensures the accuracy and reliability of user data, which is fundamental to the system's overall efficiency. These robust mechanisms provide a solid foundation for the platform, enabling it to seamlessly handle a diverse array of user interactions.

Moreover, the integration of intuitive status indicators within MitraPark plays a crucial role in enhancing user experience. Users can easily track and understand their interactions with the platform by providing clear and transparent insights into the progress of various actions, such as service requests and approvals. This level of transparency builds trust and encourages more active participation from the community, fostering a more engaged and informed user base.

Administrators are empowered with comprehensive tools to efficiently manage requests, monitor activities, and maintain accurate records. This administrative capability ensures that the platform remains responsive and adaptive to the needs of its users. Collectively, these enhancements streamline the usage of MitraPark, making it a more effective and user-

friendly tool. As MitraPark continues to evolve, it will undoubtedly contribute to more sustainable and vibrant urban living, enhancing the quality of life for all its stakeholders.

5.3 Future Recommendations

There are a few improvements and additions that can be made to the system:

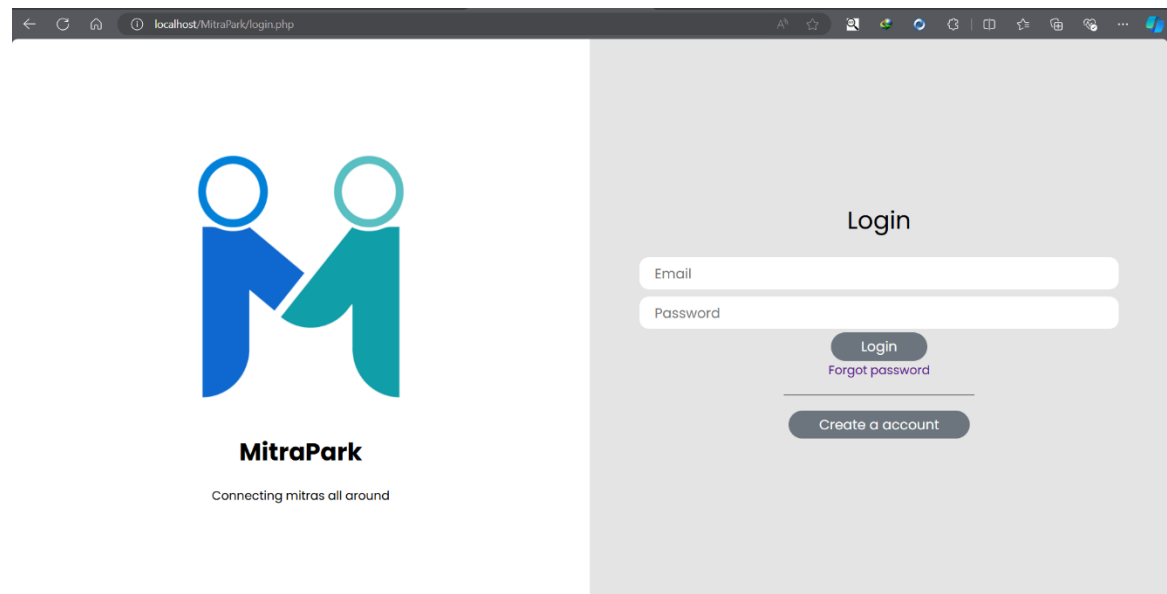
- a. Multilingual to this site can be added,
- b. Support for more multimedia like video can be added,
- c. Advanced algorithm implementation for user choice tracking and suggesting users and posts can be added,
- d. Audio calls and video can all be added,

References

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Appendices

Login page :



localhost/MitraPark/login.php

MitraPark
Connecting mitras all around

Login

Email

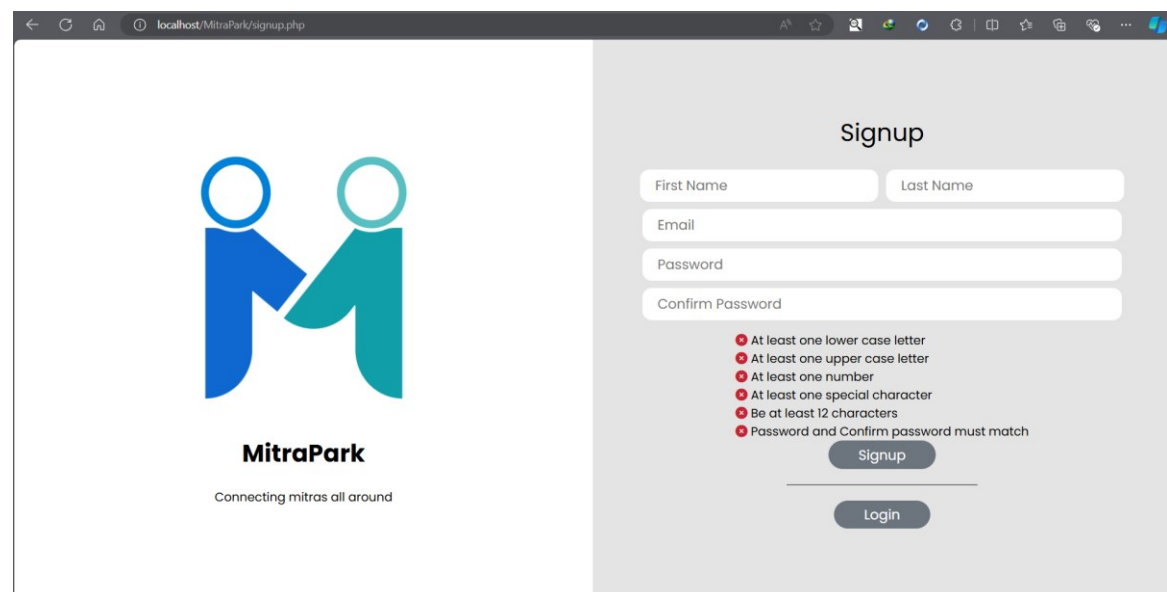
Password

Login

[Forgot password](#)

Create a account

Signup page :



localhost/MitraPark/signup.php

MitraPark
Connecting mitras all around

Signup

First Name

Last Name

Email

Password

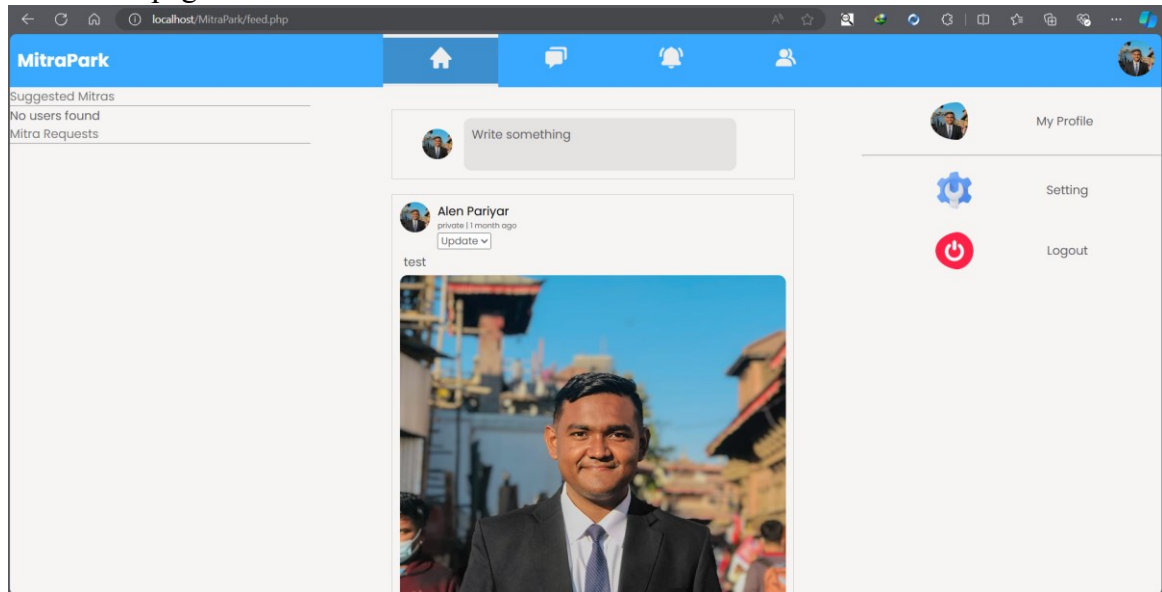
Confirm Password

- At least one lower case letter
- At least one upper case letter
- At least one number
- At least one special character
- Be at least 12 characters
- Password and Confirm password must match

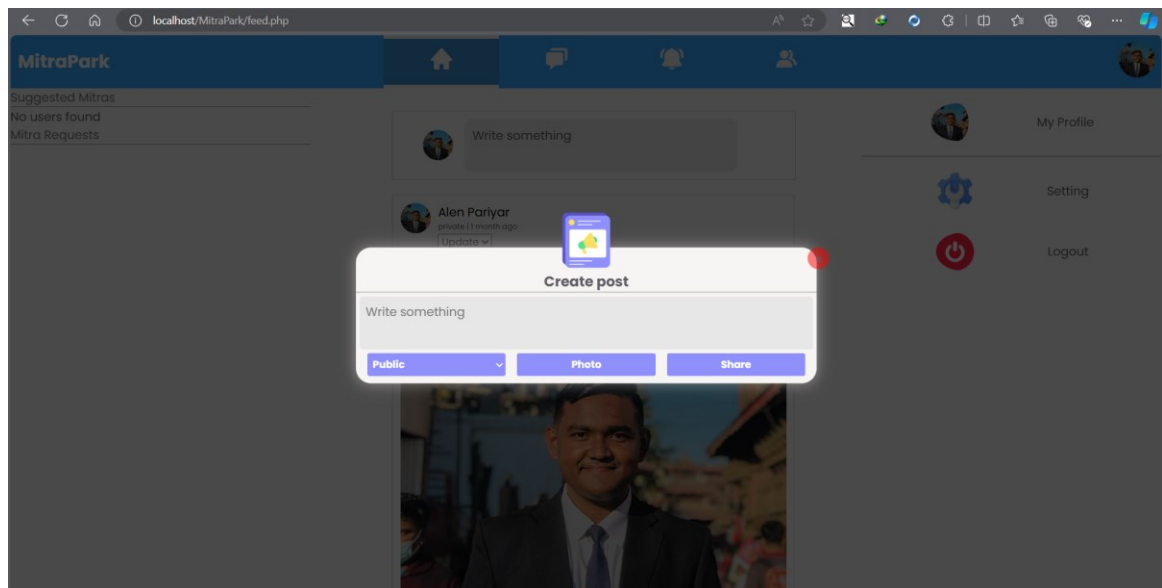
Signup

Login

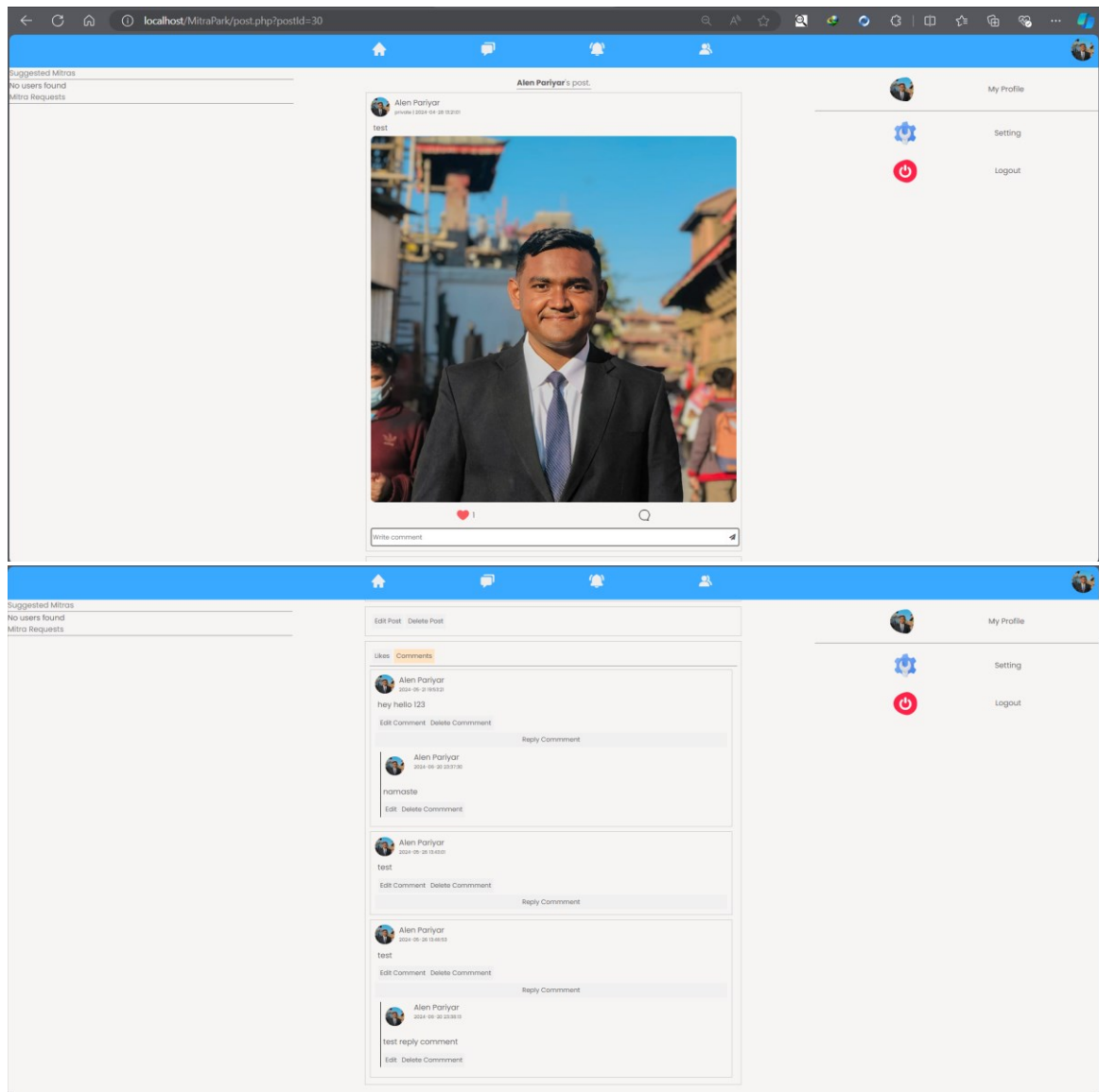
News feed page:



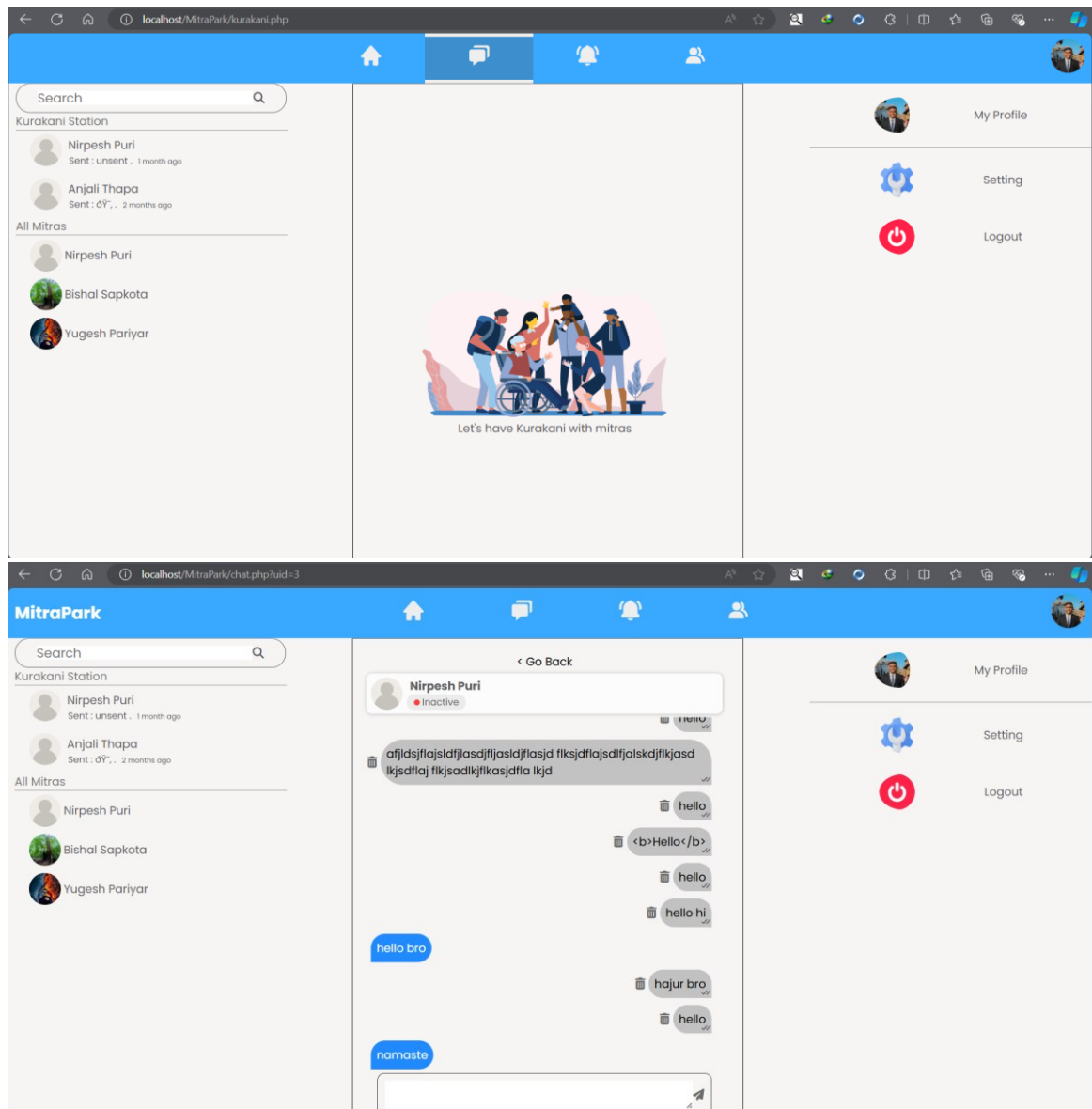
Create post :



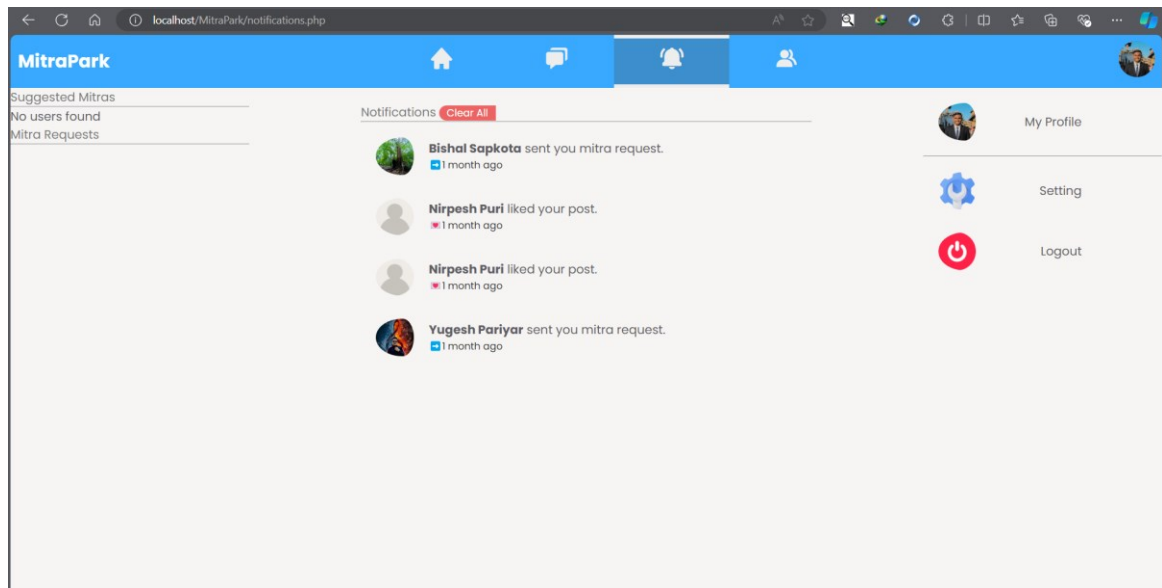
Post Page :



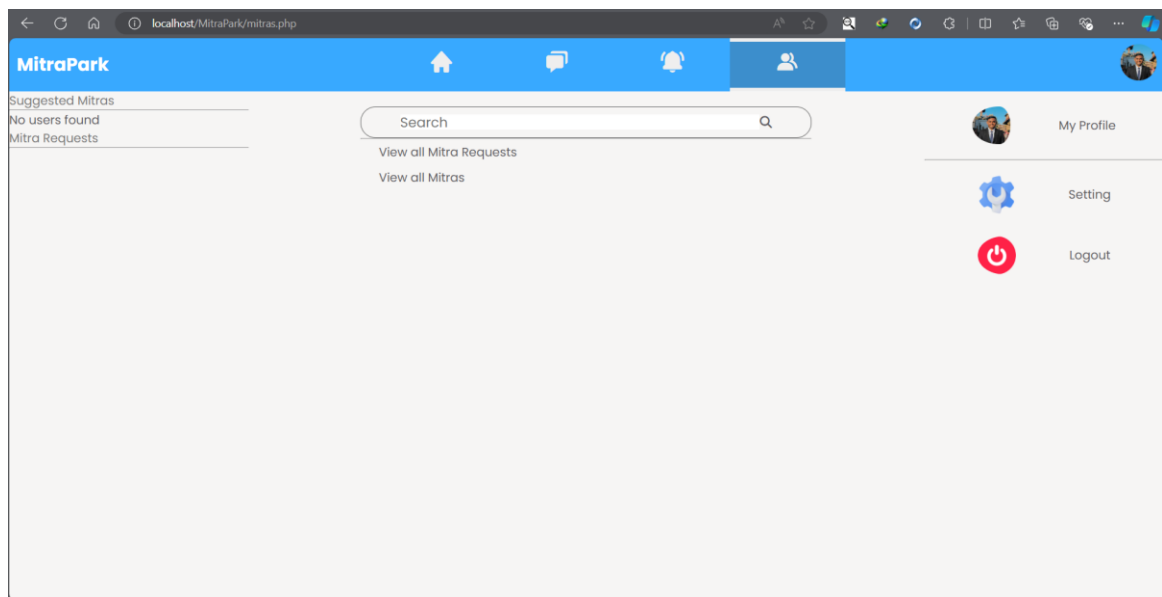
Kurakani Page :



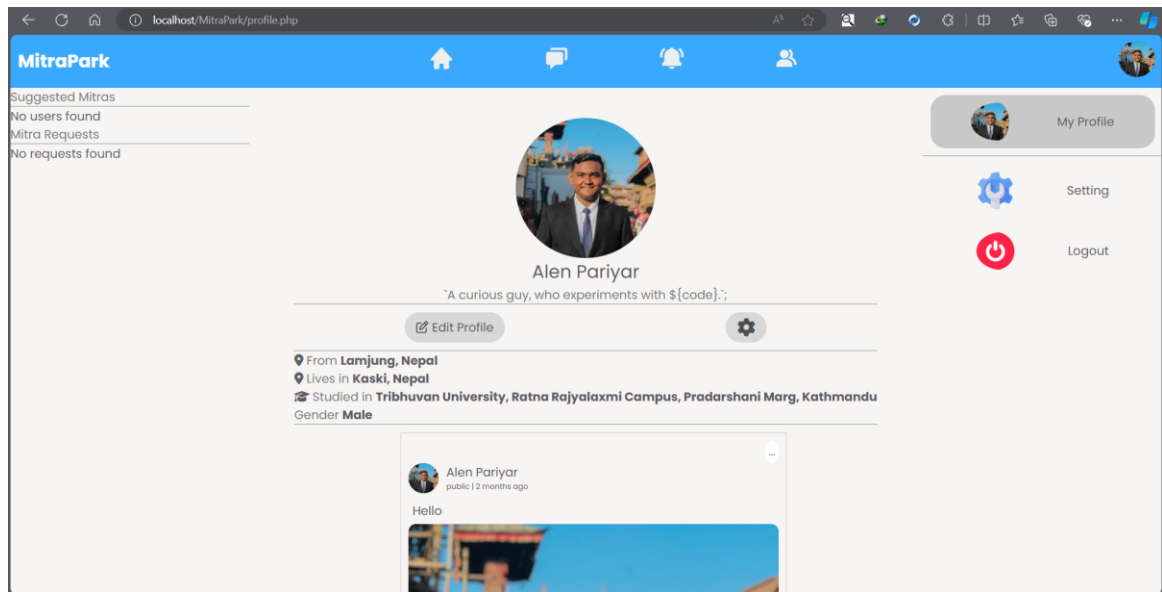
Notifications Page:



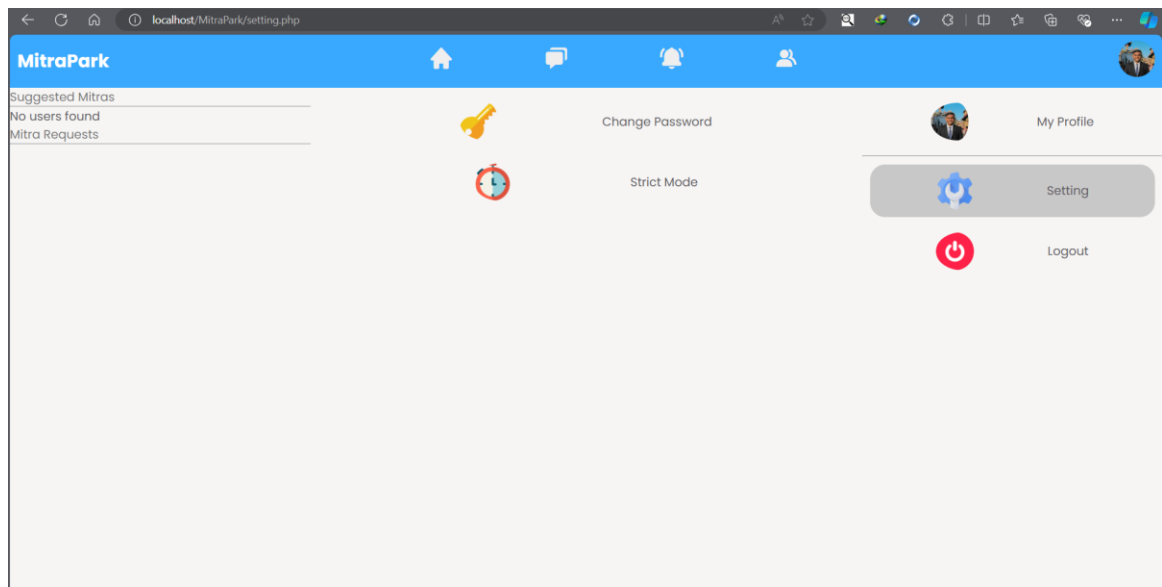
Mitras Page:



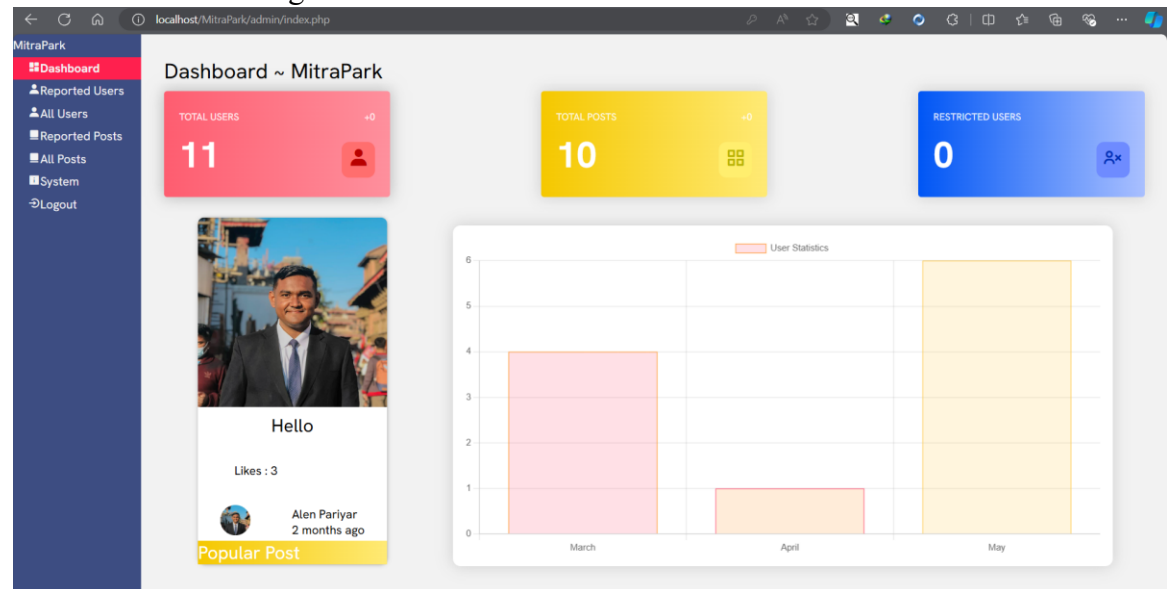
User Profile Page:



Setting Page:



Admin Dashboard Page:



Reported Users Page:

Reported Users ~ MitraPark

TOTAL REPORTED USERS: 3
[View records](#)

TOTAL RESTRICTED USERS: 0
[View records](#)

Report Id	UID	Name	Report Description	Operation
202	3	Nirpesh Puri	this is bad user	Delete Restrict View
206	4	Anjali Thapa	bad user	Delete Restrict View
209	1	Alen Pariyar	this is fake id	Delete Restrict View
210	1	Alen Pariyar	not a good user	Delete Restrict View
211	1	Alen Pariyar	not a good user	Delete Restrict View
212	1	Alen Pariyar	bad user	Delete Restrict View
213	1	Alen Pariyar	bad user	Delete Restrict View
214	1	Alen Pariyar	fake id	Delete Restrict View

All Users Page:

The screenshot shows the 'All Users' page of the MitraPark admin dashboard. The left sidebar contains navigation links: Dashboard, Reported Users, All Users (highlighted), Reported Posts, All Posts, System, and Logout. The main content area is titled 'All Users ~ MitraPark' and includes a search bar labeled 'Search by Name, UID or Email'. Below the search bar is a table with columns: UID, Name, Profile Picture, Status, and Operation. The table lists 10 users, with the first two rows highlighted. The 'Operation' column for each row contains 'Restrict' and 'View' buttons. At the bottom of the table, there is a pagination bar showing '1' and '2'.

UID	Name	Profile Picture	Status	Operation
1	Alen Pariyar		active	Restrict View
3	Nirpesh Puri		active	Restrict View
4	Anjali Thapa		active	Restrict View
5	Anurodh Aryal		active	Restrict View
6	Bishal Sapkota		active	Restrict View
8	Asmita Tharu		active	Restrict View
9	Alen Pariyar		active	Restrict View
10	Anurodh Aryal		active	Restrict View

Reported Posts Page:

The screenshot shows the 'Reported Posts' page of the MitraPark admin dashboard. The left sidebar contains navigation links: Dashboard, Reported Users, All Users, Reported Posts (highlighted), All Posts, System, and Logout. The main content area is titled 'Reported Posts ~ MitraPark'. It features two summary boxes: 'TOTAL REPORTED POSTS - Unique' with a value of 1 and a 'View records' link, and 'TOTAL RESTRICTED POSTS - Unique' with a value of 0 and a 'View records' link. Below these boxes is a table with columns: Report ID, Post ID, Report Description, and Operation. The table lists one report with Report ID 204, Post ID 25, and the description 'this post contain some fake news'. The 'Operation' column for this row contains 'Delete', 'Restrict', and 'View' buttons. At the bottom of the table, there is a pagination bar showing '1'.

Report ID	Post ID	Report Description	Operation
204	25	this post contain some fake news	Delete Restrict View

All Posts Page:

localhost/MitraPark/admin/posts.php

All Posts ~ MitraPark

Search by Author, Post Id or Caption

Post Id	Author	Caption	Media	Status	Operation
5	Nirpesh Puri	It was a good day today.		active	Restrict View
10	Alen Pariyar	this is private post		active	Restrict View
12	Alen Pariyar	Trash		active	Restrict View
14	Nirpesh Puri	hello post		active	Restrict View
17	Alen Pariyar	<script>alert("hello");</script>		active	Restrict View
23	Alen Pariyar	Hello		active	Restrict View
25	Nirpesh Puri	Namaste hai		active	Restrict View
26	Nirpesh Puri	this is rr campus		active	Restrict View

1 2

System Page:

localhost/MitraPark/admin/system.php

System ~ MitraPark

[Change Password](#) [Update System Info](#)

System Name

MitraPark

System Logo

Maintenance Mode

Off

System Color

☒ ☐ ☐

System description

Connecting mitras all around

Update System Info :

The screenshot shows the 'Update System Info' modal in the MitraPark admin panel. The modal is titled 'Update System Info' and contains the following fields and controls:

- Name :** A text input field containing 'MitraPark'.
- Description :** A text input field containing 'Connecting mitras all around'.
- Maintenance Mode :** A toggle switch currently set to 'On'.
- System Logo :** A button labeled 'Change Logo'.
- System Color Theme :** A section with three color pickers: 'Primary Color' (blue), 'Secondary Color' (dark blue), and 'Background Color' (light blue).
- Save :** A button at the bottom of the modal.

The background of the admin panel shows the 'System ~ MitraPark' header and a sidebar with navigation links: Dashboard, Reported Users, All Users, Reported Posts, All Posts, System (highlighted), and Logout.

Change Password:

The screenshot shows the 'Change Password' modal in the MitraPark admin panel. The modal is titled 'Change Password' and contains the following fields and controls:

- Current Password :** A text input field.
- New Password :** A text input field.
- Confirm Password :** A text input field.
- Validation Errors :** A list of red error messages:
 - Valid current password
 - At least one lower case letter
 - At least one upper case letter
 - At least one number
 - At least one special character
 - Be at least 12 characters
 - Password and Confirm password must match
- Save :** A button at the bottom of the modal.

The background of the admin panel shows the 'System ~ MitraPark' header and a sidebar with navigation links: Dashboard, Reported Users, All Users, Reported Posts, All Posts, System (highlighted), and Logout.